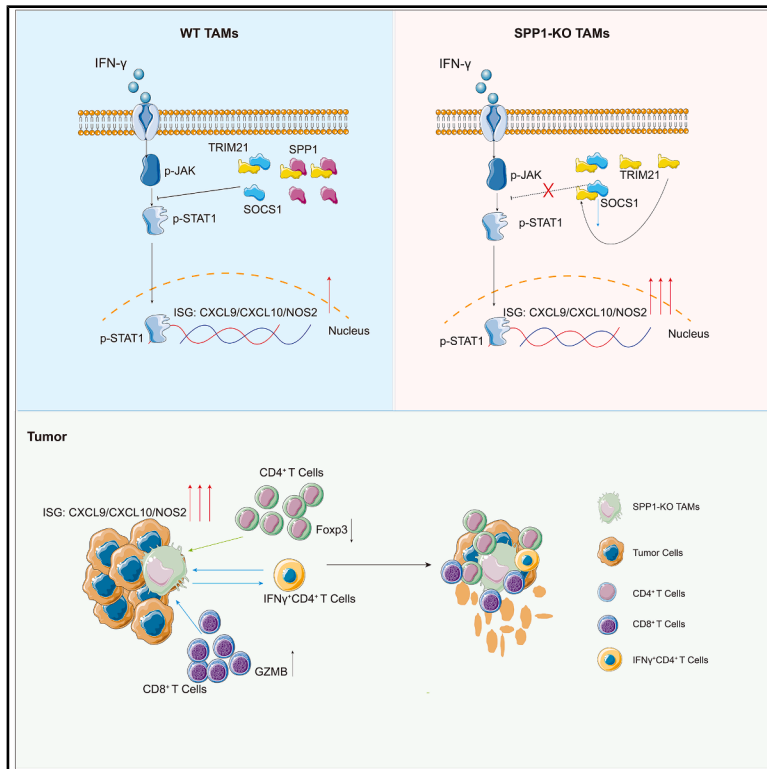


An SPP1-SOCS1 pathway constrains interferon responses in tumor-associated macrophages and shapes an immunosuppressive tumor microenvironment

Graphical abstract



Authors

Liangzhan Sun, Xiaojing Chu, Tingting Kong, ..., Sijin Cheng, Linnan Zhu, Zemin Zhang

Correspondence

zhuln@pku.edu.cn (L.Z.),
zemin@pku.edu.cn (Z.Z.)

In brief

Tumor-associated macrophages (TAMs) can limit anti-tumor immunity. Sun et al. find that SPP1⁺ TAMs are enriched in tumors across cancer types and associated with resistance to immune checkpoint blockade. Mechanistically, SPP1 interacts with TRIM21 to limit SOCS1 ubiquitination, thereby dampening IFN-γ-STAT1-ISG signaling in TAMs.

Highlights

- SPP1⁺ TAMs are enriched in tumor tissues and linked to immunotherapy resistance
- Intracellular SPP1 negatively regulates the IFN-γ-STAT1-ISG pathway in TAMs
- Intracellular SPP1 decreases SOCS1 ubiquitination by competitively binding to TRIM21
- SPP1 deletion in macrophages promotes inflammation and enhances ICB response

Article

An SPP1-SOCS1 pathway constrains interferon responses in tumor-associated macrophages and shapes an immunosuppressive tumor microenvironment

Liangzhan Sun,^{1,2,3,8} Xiaojing Chu,^{4,5,8} Tingting Kong,¹ Xirui Chen,¹ Jianing Ru,¹ Qianqian Gao,^{1,3} Wei Zhou,³ Xiliang Wang,⁶ Sijin Cheng,^{1,4} Linnan Zhu,^{7,*} and Zemin Zhang^{1,7,9,*}

¹Institute for Data-Driven Tumor Immunology, Chongqing Medical University, Chongqing 400016, China

²Peking University Shenzhen Graduate School, Peking University, Shenzhen 518055, China

³Institute of Cancer Research, Shenzhen Bay Laboratory, Shenzhen 518000, China

⁴Changping Laboratory, Beijing 102206, China

⁵Key Laboratory of Cell Proliferation and Regulation Biology, Ministry of Education, Department of Biology, College of Life Sciences, Beijing Normal University, Beijing 100875, China

⁶Analytical Biosciences Limited, Beijing 100083, China

⁷Biomedical Pioneering Innovation Center (BIOPIIC), Academy for Advanced Interdisciplinary Studies, and School of Life Sciences, Peking University, Beijing 100871, China

⁸These authors contributed equally

⁹Lead contact

*Correspondence: zhulin@pku.edu.cn (L.Z.), zemin@pku.edu.cn (Z.Z.)

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SUMMARY

Tumor-associated macrophages (TAMs) can suppress antitumor immunity and reduce responses to immune checkpoint blockade (ICB). Here, we asked how TAM programs contribute to ICB non-response. Integration of public single-cell RNA sequencing (scRNA-seq) datasets across 12 cancer types identified SPP1⁺ TAMs as a tumor-enriched macrophage subset with immunosuppressive features. TAMs from ICB non-responders across multiple tumor types exhibited higher SPP1 expression. In murine models, macrophage *Spp1* deletion suppressed tumor growth and prolonged survival and was associated with a remodeled tumor microenvironment featuring reduced T regulatory cell (Treg) frequencies, increased interferon (IFN)- γ ⁺ CD4⁺ and GZMB⁺ CD8⁺ T cells, and augmented interferon-stimulated gene (ISG) expression across immune and malignant compartments. Mechanistically, intracellular SPP1 interacted with TRIM21 to limit SOCS1 ubiquitination, stabilizing SOCS1-mediated negative feedback and dampening IFN- γ -STAT1-ISG signaling in TAMs. Consistently, SPP1 targeting enhanced the efficacy of anti-PD-L1 therapy *in vivo*. Thus, remodeling the TME via targeting the TAM SPP1-IFN- γ axis presents a therapeutic avenue for enhancing responses to ICB.

INTRODUCTION

T cell-targeting strategies, such as immune checkpoint blockade (ICB) therapies against CTLA4 and PD-1/PD-L1, as well as T cell receptor (TCR) or chimeric antigen receptor (CAR) T cell therapies, have revolutionized the immunotherapy landscape.^{1–3} However, only 15%–35% of patients with solid tumors benefit from these therapies, with the majority either failing to respond or rapidly developing resistance after an initial, transient response.^{1–3} Tumor-associated macrophages (TAMs) can eliminate cancer cells directly through phagocytosis and the secretion of cytotoxic cytokines, while also strengthening adaptive immunity by presenting antigens.^{4,5} Additionally, their capacity to infiltrate dense tumor stroma and superior tolerance to hypoxic and acidic intra-tumor environments position TAMs as up-and-coming candidates for anti-tumor immunotherapy.^{4–6}

Highlighting the transcriptomic heterogeneities within TAMs, our previous single-cell studies in multiple cancer types have reported distinct TAM subsets with different functional potentials involving phagocytosis, lymphocyte recruitment, angiogenesis, and immunotherapy resistance.^{6–9} However, these studies have not identified the key molecules and mechanisms driving the functions of TAMs in the tumor microenvironment and ICB therapy. In this study, leveraging public single-cell RNA sequencing (scRNA-seq) data from multiple cancer types, we revealed the enrichment and pro-tumor effects of SPP1⁺ TAMs in tumors from a larger cohort and prioritized SPP1 as a defining marker of this subset and further validated across multiple cancers that elevated SPP1 expression in TAMs is associated with resistance to ICB therapy. The full SPP1 transcript encodes a secreted form of the SPP1 protein, while alternative translation initiation can produce an intracellular form by excluding the signal sequence.¹⁰ Both secreted and intracellular SPP1

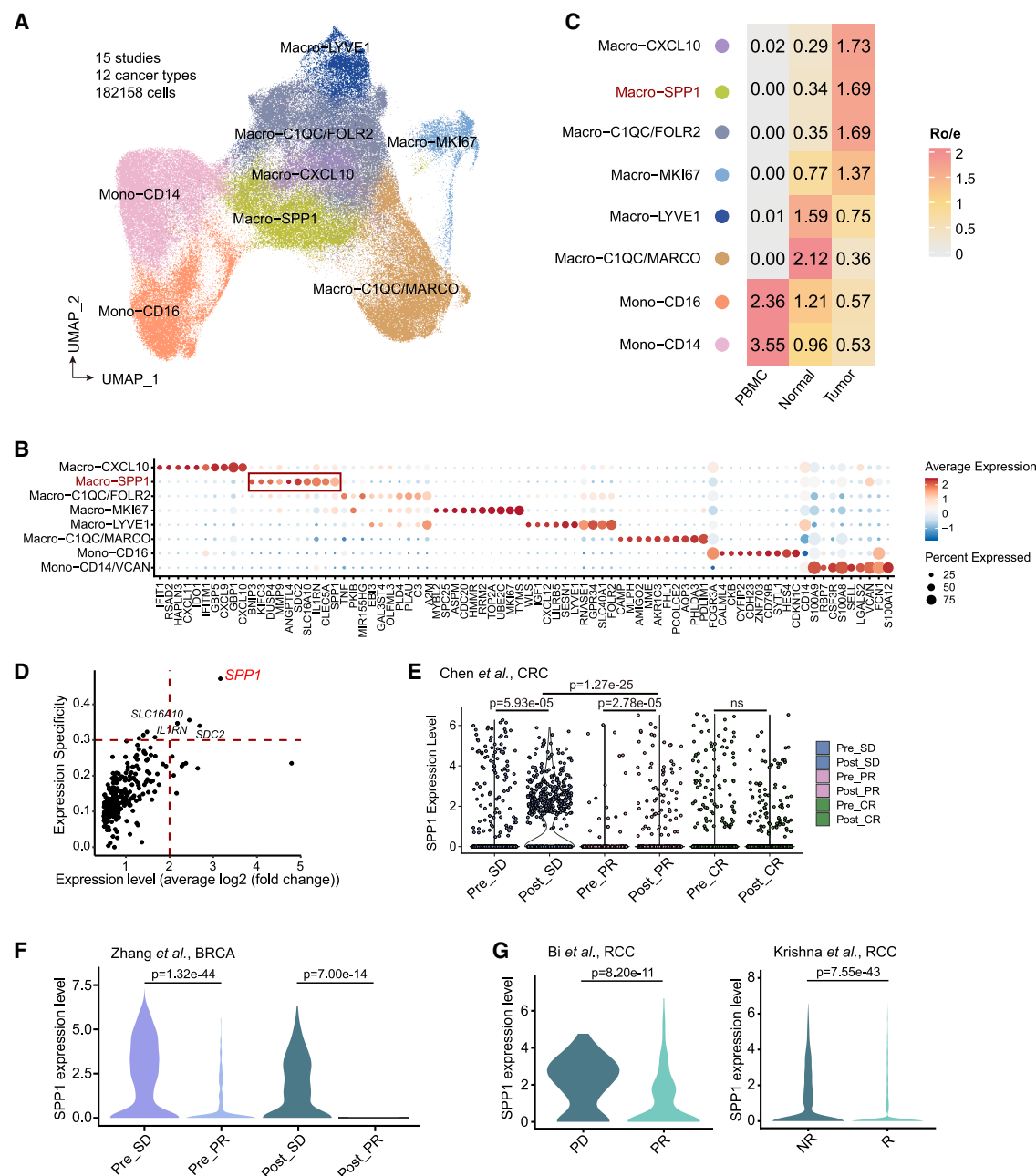


Figure 1. Integrative single-cell analysis reveals that SPP1 in TAMs is associated with pro-tumor effects

(A) Uniform manifold approximation and projection (UMAP) plot showing identified subsets of all monocytes and macrophages. $n = 182158$ cells.

(B) Dot plot showing the expression level of the top ten marker genes of each monocyte and macrophage subset. Dot size indicates the fraction of expressing cells, and dot color indicates the normalized expression level.

(C) Heatmap showing tissue preference of each monocyte and macrophage subset indicated by Ro/e.

(D) Scatterplot showing the expression difference of all marker genes of the Macro-SPP1 subset in contrast to other subsets. The x axis indicates the average log₂ fold change, and the y axis indicates expression specificity calculated by subtracting the expressing percent out of the subset from the expressing percent in the subset.

(E) Scatterplot and violin plot showing SPP1 expression in TAMs of different treatment responses at different time points of ICB therapy from colorectal cancer (CRC) patients. Pre, pre-treatment; Post, post-treatment; SD, stable disease; PR, partial response; CR, complete response. Unpaired two-sided Wilcoxon's test. $n = 565, 509, 242, 361, 739$, and 1197 cells from left to right.

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participate in macrophage polarization and immune responses, but their effects can vary and sometimes contradict based on expression context.^{11–15} SPP1⁺ TAMs have been identified in various tumor types, and our previous studies have found that these TAMs are associated with poor survival, liver metastasis, and immunotherapy resistance in colorectal cancer (CRC) patients.^{6–9} However, the specific functions of SPP1 in TAMs and the underlying mechanisms remain unclear. Moreover, although a few preclinical studies have reported that SPP1-targeting antibodies exhibit therapeutic potential, no benefit has been observed in clinical trials.^{16,17} Therefore, whether SPP1 can serve as an effective therapeutic target for cancer remains to be further investigated.

In this study, we utilized multiple cancer models to demonstrate that SPP1 knockout in TAMs significantly inhibited tumor growth and prolonged survival. Following SPP1 knockout in TAMs, inflammatory signaling within the tumor was enhanced, with increased immune cell infiltration, particularly of tumor-suppressive interferon (IFN)- γ ⁺ CD4⁺ T cells and GZMB⁺ CD8⁺ T cells, indicating a transition to a more immune-responsive environment. Mechanistically, we found that SPP1 is markedly overexpressed in TAMs, and intracellular SPP1 attenuates IFN- γ -STAT1-interferon-stimulated gene (ISG) signaling by limiting SOCS1 ubiquitination through competitive binding to TRIM21, thereby acting as a molecular shock absorber of interferon signaling. Importantly, SPP1 ablation in TAMs significantly improved the efficacy of anti-PD-L1 immunotherapy. Together, these findings suggest that SPP1 is a promising target to reverse the immunosuppressive roles of TAMs for cancer immunotherapy.

RESULTS

SPP1 in the pro-tumor role of TAMs

To comprehensively characterize the gene expression landscape of tumor-infiltrating monocytes and macrophages, we collected single-cell transcriptomic data of 135 donors from 15 publicly available datasets covering 12 cancer types (Table S1).^{6,9,13,14,18–28} Following batch correction and quality control, 182,158 single cells from tumors, adjacent normal tissues, and peripheral blood mononuclear cells (PBMCs) were kept for clustering (Figures 1A and S1A). Based on the distinct gene expression patterns, we subsequently classified monocytes and macrophages into eight subsets, including two monocyte subsets (Mono-CD16 and Mono-CD14) and six macrophage subsets characterized by the differential expression of *CXCL10*, *SPP1*, *C1QC*, *MACRO*, *LYVE1*, and *MKI67* (Figures 1A, 1B, and S1A; Table S2). To assess the tissue distribution of these monocyte and macrophage subsets, we used the ratio of observed to expected cell numbers (Ro/e)^{29,30} and revealed that CXCL10⁺ TAMs, SPP1⁺ TAMs, C1QC⁺ TAMs, and a subset of proliferating MKI67⁺ TAMs were enriched in tumor tissues compared with adjacent normal tissues and PBMCs

(Figure 1C). Notably, among these tumor-enriched cell subsets, SPP1⁺ TAMs stood out, consistent with previous studies suggesting their pro-tumor roles in CRC.^{6–9} Our integrative scRNA-seq data further confirmed that this subpopulation exhibited pro-tumor characteristics by upregulating pathways related to angiogenesis, response to hypoxia, cell-matrix adhesion, and transforming growth factor β (TGF- β) production, as well as negative regulation of adaptive immune response (Figure S1B). Furthermore, using our recently published CRC atlas,⁷ we found that SPP1⁺ TAM abundance was positively correlated with T regulatory cell (Treg) abundance and the co-inhibitory score of CD8⁺ T cells (Figures S1C and S1D). These results reinforced the immune-suppressive potential of macrophages marked with high expression of *SPP1*.

To uncover key regulators of the SPP1⁺ TAMs phenotype, we analyzed differentially expressed genes and found that SPP1 exhibited the most significant upregulation and specificity (Figure 1D). Additionally, by comparing the expression levels of *SPP1* in tumor-infiltrating macrophages before and after ICB therapy in publicly available datasets, we observed higher *SPP1* expression in TAMs from patients with poor response across multiple cancer types (Figures 1E–1G and S1E).^{7,31–34} Consistent with these findings, pan-cancer bulk RNA sequencing (RNA-seq) data revealed significantly elevated *SPP1* expression in tumors compared with adjacent normal tissues across 16 cancer types, including colon adenocarcinoma (COAD) and rectal adenocarcinoma (READ) (Figure S1F). Collectively, these results prompted us to investigate further the functional roles of SPP1 in TAMs and its impact on the tumor microenvironment, aiming to better understand its roles in tumor progression and immunotherapy resistance.

Knockout SPP1 in TAMs delays tumor growth

To investigate the role of SPP1 in cancer, we generated myeloid-specific SPP1 knockout (Spp1-KO) mice and established a subcutaneous tumor model using the MC38 CRC and LLC1 lung cancer cell line. Myeloid Spp1 deletion significantly inhibited tumor growth (Figures 2A, S2A, and S2B). Given reports of SPP1⁺ neutrophils with potential pro-tumor activity,^{35,36} we sought to determine whether the phenotype reflects SPP1 loss in TAMs or in neutrophils. However, in this setting, SPP1 expression in neutrophils is exceedingly low, and SPP1⁺ neutrophils are virtually undetectable (Figures S2C and S2D). Additionally, we performed *in vivo* macrophage depletion and repeated subcutaneous implantation (Figure S2E). The antitumor effect was abolished, demonstrating that it depends on TAMs rather than neutrophils (Figure S2F).

Consistent with the observation in the subcutaneous tumor model, tumor progression was also significantly delayed in a peritoneal tumor model established by injecting luciferase-labeled MC38 cells into the peritoneal cavity, resulting in reduced tumor burden and markedly prolonged survival in

(F) Violin plots showing SPP1 expression in TAMs in different treatment responses at different time points of ICB therapy from breast invasive carcinoma (BRCA) patients. Pre, pre-treatment; Post, post-treatment; SD, stable disease; PR, partial response. Unpaired two-sided Wilcoxon's test. $n = 3373, 443, 843$, and 85 cells from left to right.

(G) Violin plots showing SPP1 expression in TAMs of different treatment responses of ICB therapy from renal cell carcinoma (RCC) patients. PD, progressive disease; PR, partial response; NR, non-responder; R, responder. Unpaired two-sided Wilcoxon's test. $n = 107, 754, 7872$, and 4195 cells from left to right. See also Figure S1.

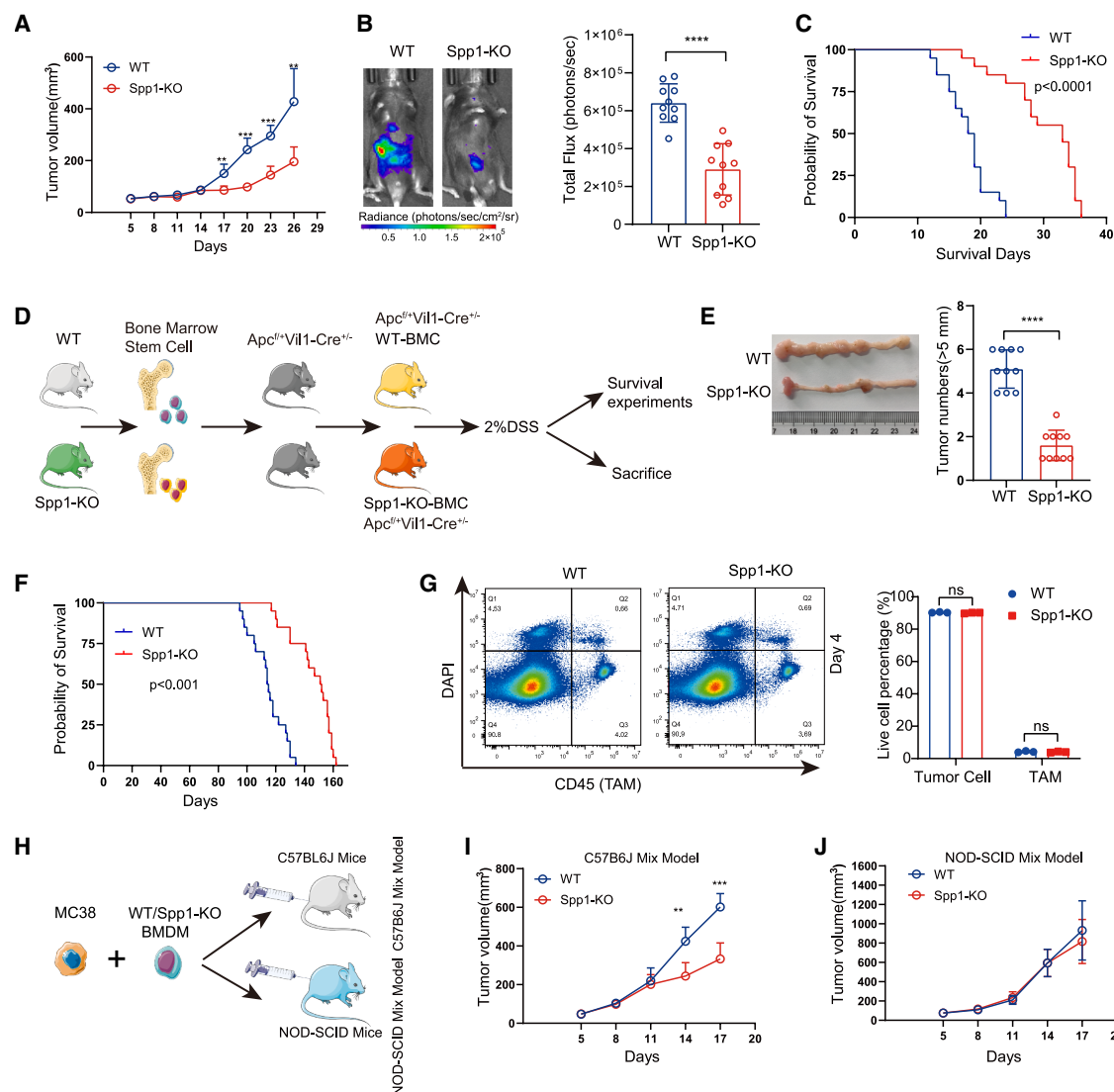


Figure 2. SPP1 knockout in TAMs delays tumor progression in a lymphocyte-dependent manner

(A) Tumor growth curves after 2×10^6 MC38 cells subcutaneously transplanted into WT/Spp1-KO mice ($n = 6$ per group).
 (B) Bioluminescence analysis of the abdominal tumor burden of the indicated mice 14 days after intraperitoneal injection of 2×10^6 MC38-luciferase cells. The representative picture (right) and the statistical result (left) are shown ($n = 10$ per group).
 (C) Survival analysis of the indicated mice after intraperitoneal injection of 2×10^6 MC38-luciferase cells. Survival analysis was performed using the log-rank (Mantel-Cox) test ($n = 20$ per group).
 (D–F) Strategy for generating orthotopic CRC model (D). The representative picture (right) and the statistical result of tumor number (left) (E) and survival analysis of indicated groups (F). Survival analysis was performed using the log-rank (Mantel-Cox) test (tumor number: $n = 10$ per group; survival analysis: $n = 20$ per group).
 (G) Representative fluorescence-activated cell sorting (FACS) analysis of tumor and TAMs cells mixed 1:1 and co-cultured for 4 days. Representative picture (right) and the statistical result (left) ($n = 3$ per group).
 (H) The schematic diagram for generating the NOD-SCID Mix and C57BL/6J Mix model.
 (I and J) Tumor growth curves after 2×10^6 MC38 mixed with 2×10^6 WT/Spp1-KO BMDM cells and transplanted into immunocompetent C57BL/6J (I) or NOD-SCID (J) mice (NOD-SCID Mix model: $n = 5$ per group; C57BL/6J Mix model: $n = 7$ per group).
 Data are shown as mean $\pm$ SEM; ns: non-significant. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$, unpaired Student's t test, two-tailed (A, B, E, G, I, and J). See also Figure S2.

Spp1-KO mice compared with wild-type (WT) controls (Figures 2B and 2C). We next evaluated this in a colorectal orthotopic model that better recapitulates human disease: we generated bone-marrow chimeric (BMC) mice by transplanting WT or Spp1-KO marrow into APC/DSS-induced CRC recipients, as

described previously³⁷ (Figure 2D). After SPP1 knockout, the number of large tumors (>5 mm) significantly decreased (Figure 2E), although the total number of colorectal tumors remained unchanged (Figure S2G), and survival was significantly extended (Figures 2F and S2H). Together, these results

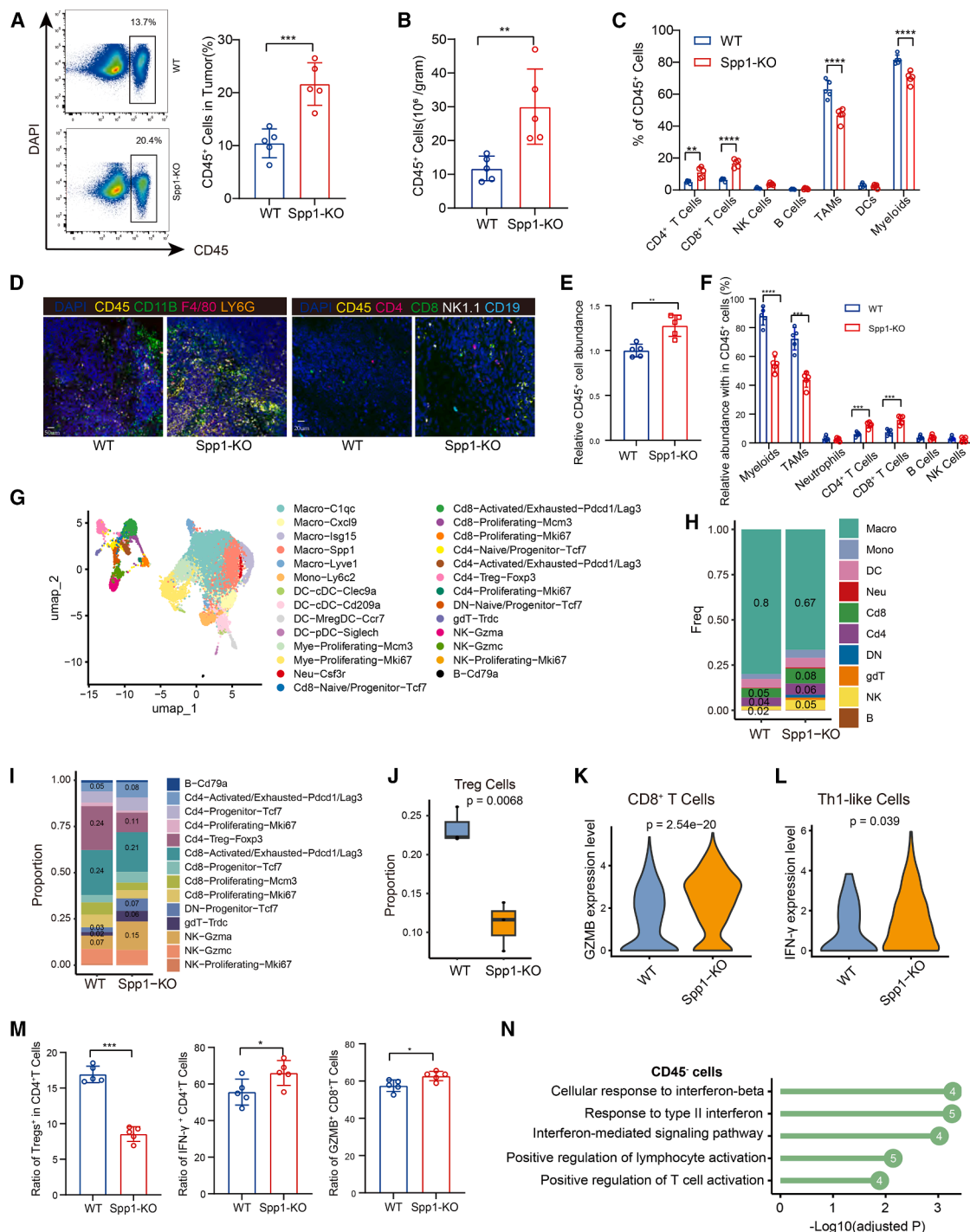


Figure 3. Knocking out SPP1 in TAMs reshapes the tumor microenvironment

(A) FACS analysis of the frequency of immune cell infiltration in the indicated tumor. Representative picture (left) and statistical result (right) ($n = 5$ per group). (B) FACS analysis of absolute immune cell number per gram of tumor ($n = 5$ per group). (C) FACS analysis of the frequency of CD4⁺ T cells, CD8⁺ T cells, NK cells, B cells, TAMs, and myeloid cells as a percentage of total immune cells ($n = 5$ per group). (D–F) mIF of murine tumors. Representative images (D), relative abundance of CD45⁺ immune cells (E), and fractional composition of major immune lineages within the CD45⁺ compartment (F) in WT versus Spp1-KO ($n = 5$ per group). (G and H) scRNA-seq analysis of the CD45⁺ immune cells in indicated tumors ($n = 3$ per group). UMAP plot of CD45⁺ immune cell clusters from merged conditions, $n = 17659$ cells (G). Stacked columns show the major lineages' proportions in indicated tumors (H). (I) Stacked columns show the cluster proportions of lymphoid cells in indicated tumors. (J) Boxplot showing the proportion of regulatory T cells (Tregs) in the indicated tumors ($n = 3$). Unpaired two-sided t test.

indicated that SPP1 knockout in TAMs inhibited tumor progression and prolonged survival in multiple cancer models.

To examine whether SPP1 in TAMs could directly affect cancer cells, we first cultured CT26 and MC38 CRC cells with TAM-conditioned media. As shown by the cancer cell clone formation assay, SPP1 deficiency did not significantly affect cancer cell proliferation (Figure S2I). Next, by co-culturing bone-marrow-derived macrophages (BMDMs) with MC38 cells, we found cancer cell proliferation was not altered when SPP1 was knocked out (Figures 2G and S2J), indicating that SPP1 in TAMs did not influence cancer cell growth directly.

To further investigate the necessity of lymphocytes in the anti-tumor effect of Spp1-KO in TAMs, we co-implanted WT or Spp1-KO BMDMs with MC38 cells into either immunocompetent C57BL/6J mice (C57BL/6J Mix model) or immunodeficient non-obese diabetic (NOD)-severe combined immunodeficiency (SCID) mice (NOD-SCID Mix model), which lacked T, B, and functional natural killer (NK) cells (Figure 2H). In contrast to the significant tumor growth inhibition observed in the C57BL/6J Mix model, this effect was completely abrogated in the NOD-SCID Mix model (Figures 2I, 2J, S2K, and S2L). These results indicated that tumor growth inhibition caused by SPP1 knockout in TAMs requires the synergistic action of lymphocytes, while the absence of SPP1 in TAMs alone does not directly affect tumor growth.

SPP1 knockout in TAMs enhances immune activity

To dissect the potential cellular and molecular mechanisms underlying the anti-tumor effects of SPP1 knockout in TAMs, we examined the immune cell infiltration by flow cytometry and multicolor immunofluorescence (mIF) staining. SPP1 deletion led to a marked increase in immune cell infiltration (Figures 3A–3F). Further analysis revealed a higher proportion of CD4⁺ and CD8⁺ T cells, along with a reduced proportion of myeloid cells, especially TAMs in the Spp1-KO group (Figures 3C and 3F).

To pinpoint cell populations involved, we performed scRNA-seq to analyze CD45⁺ and CD45[−] cell populations within the tumor microenvironment. Within the CD45⁺ compartment, we resolved 10 major lineages comprising 27 subpopulations based on canonical markers (Figures 3G and S3A). Because of technical constraints, scRNA-seq may underrepresent MDSCs and neutrophils,³⁸ but for other immune subsets, it is concordant with flow cytometry and mIF results, notably showing higher CD4⁺/CD8⁺ T cell proportions and lower TAM proportions in the Spp1-KO group (Figures 3C, 3F, and 3H). Pathway enrichment analysis of CD45⁺ cells revealed activation of type I and II interferon responses following SPP1 knockout (Figure S3B), suggesting an enhanced immune response. Additionally, through the analysis of lymphoid cells obtained from scRNA-seq, we identified 1 B, 9 T, and 3 NK cell subsets (Figures 3G and 3I). Notably, the proportion of Tregs was reduced in the

Spp1-KO group (Figure 3J). In CD8⁺ T cells, we observed increased expression of GZMB after SPP1 knockout in TAMs (Figure 3K). Additionally, IFN- γ expression was markedly elevated in Th1-like cells within the Spp1-KO group (Figure 3L). Cell-cell interaction analysis indicated that TAMs exhibited increased interactions with both CD4⁺ and CD8⁺ T cells after SPP1 knockout (Figure S3C). Flow cytometry analysis confirmed these results, showing a reduction in Tregs and an increase in both IFN- γ ⁺ CD4⁺ T cells and GZMB⁺ CD8⁺ T cells within the tumors of the Spp1-KO group (Figure 3M).

In the CD45[−] cell population, we identified malignant cells, endothelial cells (ECs), and cancer-associated fibroblasts (CAFs) based on the distinctive expression of known marker genes (Figures S3D and S3E). Significant upregulation of type I and II interferon responses was observed after SPP1 knockout in TAMs (Figure 3N). Specifically, the ISG and major histocompatibility complex class I (MHC class I) signatures, both associated with the ICB response,^{39,40} were markedly upregulated in malignant cells in the Spp1-KO group (Figure S3F). These data showed that SPP1 knockout in TAMs enhanced immune cell infiltration, boosted type I and II interferon responses, and upregulated ISG and MHC class I signatures, indicating an increased immune activation state, which may improve the tumor's responsiveness to immunotherapy.

SPP1 regulates ISG expression in TAMs

Next, we aimed to investigate the effects of SPP1 knockout in TAMs and how these changes influence the tumor microenvironment. Our scRNA-seq analysis identified four major myeloid cell compartments: macrophages, monocytes, neutrophils, and dendritic cells (DCs) (Figure 4A). Within the macrophage compartment, we resolved five subsets: C1qc⁺, Cxcl9⁺, Spp1⁺, Isg15⁺, and Lyve1⁺ (Figure 4A). Although SPP1 deletion did not notably change the overall proportion of TAMs, including the Spp1⁺ TAMs subset, within the myeloid compartment, gene expression analysis revealed a marked increase in ISGs, such as *Cxcl9*, *Cxcl10*, *Isg15*, *Nos2*, *Stat1*, and *Irf1* (Figures 4B, 4C, and S4A–S4C), which are associated with T cell recruitment and tumor cell killing. Pathway enrichment analysis showed that SPP1 deletion in TAMs led to an enrichment of inflammatory response, interferon-gamma, and interferon-alpha response pathways (Figures 4D and S4D). Additionally, alignment with public macrophage polarization and signature data indicated a shift toward M1-like and tumor-suppressive signatures after SPP1 deletion (Figures 4E, 4F, and S4E).^{9,41} Interestingly, unlike the *in vivo* TAMs analyzed through scRNA-seq, our bulk RNA-seq analysis of TAMs obtained from *in vitro* co-culture (Figures S4F and S4G) revealed that SPP1 deletion had no significant impact on interferon (IFN) responses or ISG expression (Figures S4H and S4I). The expression of ISGs, such as *Cxcl9*, *Cxcl10*, *Isg15*, and *Nos2*, did not show significant changes either

(K and L) Violin plots showing the expression level of GZMB in CD8⁺ T cells, $n = 517$ (WT) and 617 (Spp1-KO) cells (K), and IFN- γ in Th1-like cells, $n = 59$ (WT) and 142 (Spp1-KO) cells (L), unpaired two-sided Wilcoxon's test.

(M) FACS analysis of the frequency of Tregs, IFN- γ ⁺ CD4⁺ T cells and GZMB⁺ CD8⁺ T cells in the indicated tumor ($n = 5$ per group).

(N) Lollipop plot showing the enriched Hallmark pathways in the upregulated genes in CD45[−] cells in the Spp1-KO group compared with the WT group. Overrepresentation analysis (BH adjustment).

Data are shown as mean $\pm$ SEM, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$ by unpaired Student's t test, two-tailed (A–C, E, F, J, and M).

See also Figure S3.

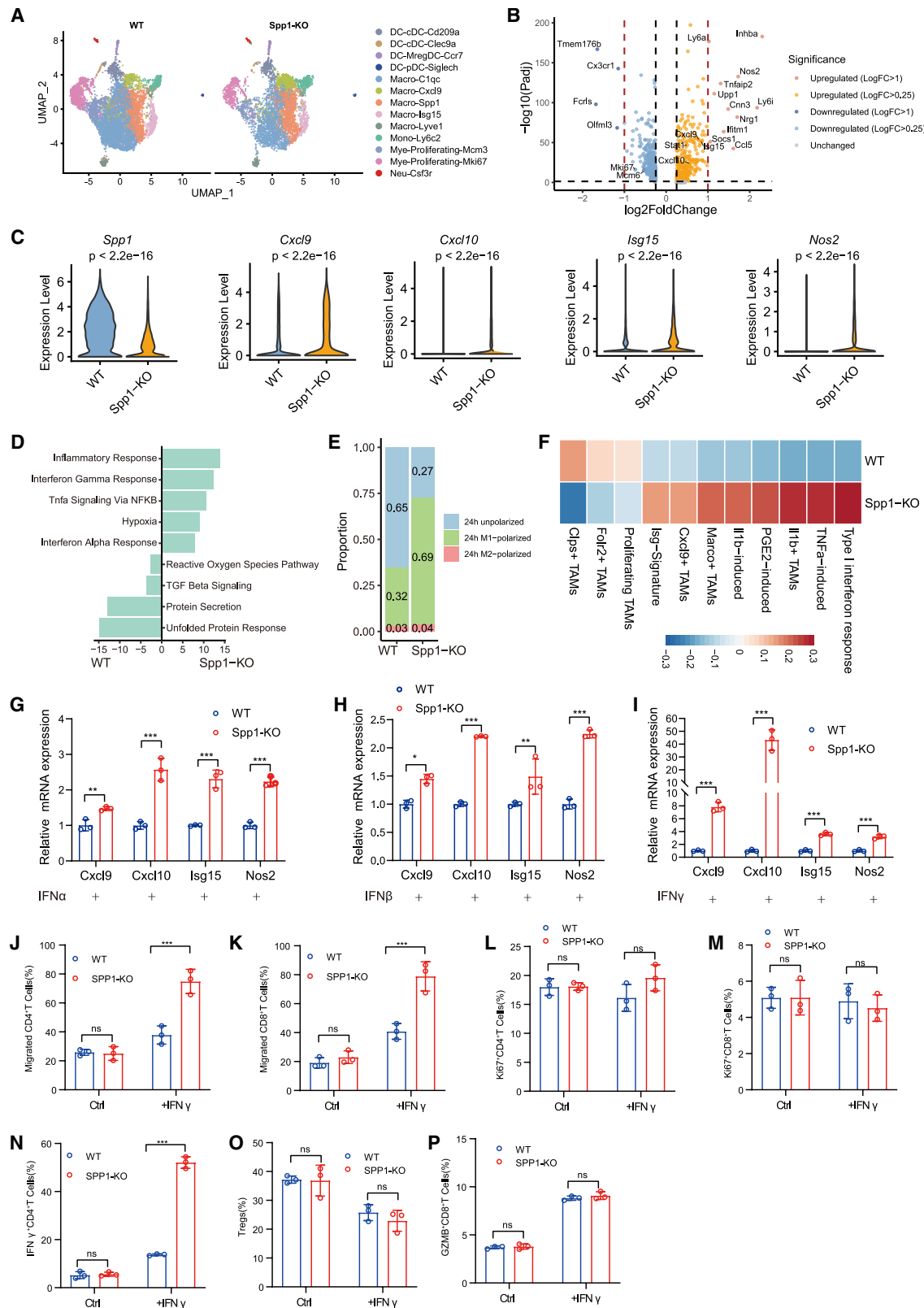


Figure 4. The role of SPP1 in TAMs

(A) Global representation of myeloid cells in the single-cell analysis, including UMAP plots denoting cell types (left) and genotypes of the tumors of origins (right) ($n = 14699$ cells).

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(Figures S4J and S4K). Since *in vivo* scRNA-seq analysis indicated that SPP1 deletion in TAMs enhanced their response to both type I and type II interferons (Figure 4D), it was noteworthy that *in vitro*-cultured TAMs lacked the stimulation from endogenous IFN present *in vivo*. Thus, we treated *in vitro*-induced TAMs with type I or type II interferons to assess the impact of SPP1 knockout on their interferon response. After interferon activation, Spp1-KO TAMs displayed increased expression of *Cxcl9*, *Cxcl10*, *Isg15*, and *Nos2* (Figures 4G–4I), while ISG expression remained unchanged in the absence of interferon treatment (Figures S4J and S4K). These findings suggested that SPP1 knockout in TAMs enhanced ISG expression and drove a transition toward an anti-tumor phenotype upon interferon activation.

To define how SPP1 loss in TAMs reshapes the tumor microenvironment, we functionally profiled T cell trafficking and fate in co-cultures with WT or Spp1-KO TAMs. In Transwell assays, SPP1-deficient TAMs recruited more CD4⁺ and CD8⁺ T cells under IFN- γ stimulation (Figures 4J and 4K), whereas T cell proliferation (Ki67⁺ CD4⁺/CD8⁺) was not significantly altered (Figures 4L and 4M). Regarding effector differentiation/function, Spp1-KO TAMs promoted Th1-like differentiation of naive CD4⁺ T cells (IFN- γ ⁺) without increasing Treg (CD25⁺FOXP3⁺) generation (Figures 4N and 4O), and GZMB⁺ CD8⁺ T cells were unchanged (Figure 4P). Consistent with these *in vitro* readouts, our scRNA-seq data showed greater lymphocyte infiltration and higher frequencies of Th1-like cells in Spp1-KO tumors. Collectively, these data indicate that SPP1 deletion in TAMs primarily enhances lymphocyte recruitment and biases CD4⁺ T cells toward a Th1-like state upon IFN- γ stimulation, providing a mechanistic link between SPP1 loss in TAMs and a more immunostimulatory TME.

Intracellular SPP1 modulates SOCS1 abundance

Subsequently, we sought to investigate how SPP1 influenced the expression of ISG under interferon stimulation in TAMs. The IFN-STAT1-ISG pathway is a well-established mechanism by which interferons regulate ISG expression. Western blot analysis demonstrated that Stat1 expression and activation were elevated in Spp1-KO TAMs upon interferon stimulation (Figure 5A). Since IFN- γ predominated over other interferons in our tumor model, we used only IFN- γ for subsequent *in vitro* experiments (Figures S5A–S5D). We observed that ISG expression was elevated in the Spp1-KO TAMs following treatment with the same dose of IFN (Figures 4G–4I). Therefore, we speculated that

SPP1 might influence the process of interferon reception or the downstream intracellular signal transmission. First, we evaluated the expression of IFN receptors in TAMs and observed no differences, suggesting that SPP1 does not influence interferon reception by regulating the expression of these receptors (Figure 5B). Considering that SPP1 could be secreted extracellularly and is a key component of the extracellular matrix, we tested whether extracellular matrix SPP1 could affect interferon binding to its receptors. Supplementing Spp1-KO TAMs with recombinant SPP1 protein to mimic its secreted form failed to rescue the elevated ISG expression and STAT1 activation (Figures 5A and 5C). Recombinant Spp1 protein was biologically active, as analysis of bulk RNA-seq data from our *in vitro* co-culture model demonstrated that Spp1 knockout in TAMs reduced *Mmp13* expression in cancer cells, while the addition of recombinant Spp1 restored *Mmp13* levels (Figures S5E and S5F). Collectively, these results suggest that SPP1 does not affect the interferon signal reception.

Next, we hypothesized that SPP1 might regulate ISG expression by participating in the intracellular signaling downstream of IFN- γ . To test this, we synthesized intracellular (iSpp1) and secretory Spp1 (sSpp1) mRNA via *in vitro* transcription and delivered them into Spp1-KO TAMs. We found that intracellular Spp1, but not the secretory Spp1, restored ISG levels resulting from Spp1 deletion (Figures 5D and S5G). To investigate how intracellular Spp1 influences IFN- γ signal transduction, we supplemented Spp1-KO TAMs with intracellular Spp1-3XFlag mRNA, then activated such TAMs with IFN- γ , and conducted immunoprecipitation. Mass spectrometry analysis of the precipitated proteins revealed a strong interaction between the Spp1 protein and the E3 ubiquitin ligase Trim21 (Figures 5E and S5H). Pathway enrichment analysis of the precipitated proteins also showed that these proteins were related to the ubiquitination process (Figures S5I and S5J). Interestingly, TRIM21 has been reported to interact with the Janus kinase (JAK)-STAT negative regulator suppressor of cytokine signaling 3 (SOCS3) and promote its degradation through the ubiquitin-proteasome system.⁴² SOCS1, structurally similar to SOCS3 and a crucial negative regulator of the IFN- γ -STAT1-ISGs pathway, exhibited elevated expression in Spp1-KO TAMs (Figure S5K). To examine whether Trim21 also interacted with Socs1, we performed immunoprecipitation experiments under IFN- γ stimulation and discovered that Trim21 indeed interacted with Socs1 (Figure 5F). These findings demonstrated that Spp1, Socs1, and Socs3 were all

(B) Volcano plot showing differentially expressed genes (DEGs) in comparing WT and Spp1-KO TAMs in scRNA-seq analysis.

(C) Violin plots showing the expression level of the indicated genes in WT/Spp1-KO TAMs ($n = 8140$ (WT) and 4978 (Spp1-KO) cells). Unpaired two-sided Wilcoxon's test. Bonferroni correction.

(D) Barplot showing the Gene Ontology (GO) pathways differentially activated in TAMs from different groups, and the x axis indicates the t-statistic from the unpaired two-sided t test.

(E) Stacked bar plots showing the predicted cell states of TAMs of WT/Spp1-KO groups.

(F) Heat map showing the gene signature scores of the WT/Spp1-KO TAMs.

(G–I) qPCR tests measuring the expression levels of indicated ISG treated with IFN- α (G), IFN- β (H), and IFN- γ (I).

(J and K) Percent of migrated CD4⁺ (J) and CD8⁺ (K) T cells in response to WT vs. Spp1-KO TAMs under Ctrl or +IFN- γ conditions ($n = 3$ per group).

(L and M) FACS analysis of the frequency of Ki67⁺ CD4⁺ (L) and Ki67⁺ CD8⁺ (M) T cells after co-culture with control or IFN- γ -activated WT and Spp1-KO TAMs ($n = 3$ per group).

(N–P) FACS analysis of the frequency of IFN- γ ⁺ CD4⁺ T cells (N), Tregs (O), and GZMB⁺ CD8⁺ T cells (P) after naive CD4⁺ T cell/naive CD8⁺ T cell co-culture with control or IFN- γ -activated WT and Spp1-KO TAMs ($n = 3$ per group).

Data are shown as mean $\pm$ SEM, ** $p < 0.01$, *** $p < 0.001$ by unpaired Student's t test, two-tailed (G–P).

See also Figure S4.

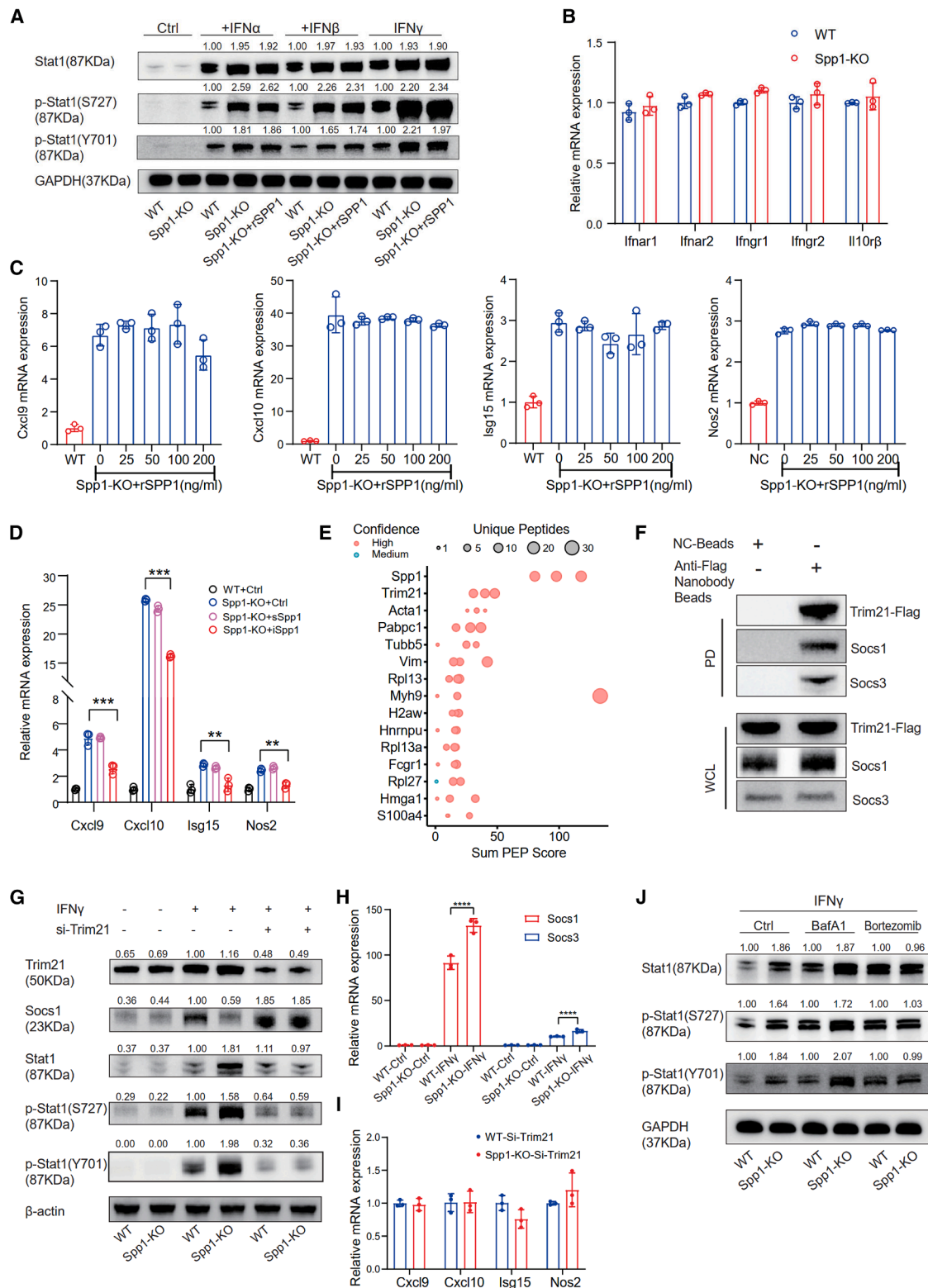


Figure 5. The mechanism of SPP1 regulating IFN- γ -STAT1-ISG signaling in TAMs

(A) Western blot analysis of WT/Spp1-KO TAMs cultured with complete growth media (10% FBS) alone and Spp1-KO TAMs with 100 ng/ml recombinant Spp1(rSpp1) protein after being treated with PBS (Ctrl), IFN- α (25 ng/ μ l), IFN- β (25 ng/ μ l), and IFN- γ (25 ng/ μ l) for 48 h. The expression of Stat1, p-Stat1(S727), and p-Stat1(Y701) under the indicated treatment was standardized against the corresponding WT group. Using β -actin as a loading control. Immunoblots are representative of two independent experiments.

(legend continued on next page)

substrates for Trim21. Therefore, we speculated that SPP1 might influence the protein abundance of SOCS1 by binding to TRIM21 competitively. In WT TAMs, both SOCS1 mRNA and protein expression were upregulated following IFN- γ treatment (Figures 5G and 5H). By contrast, although SOCS1 mRNA levels increased in Spp1-KO TAMs, protein abundance remained unchanged under IFN- γ stimulation, resulting in enhanced downstream Stat1 activation (Figures 5G and 5H). To further verify that the differences in Socs1 protein abundance between IFN- γ -treated WT and Spp1-KO TAMs were attributable to Trim21, we silenced Trim21 and observed an increase in SOCS1 protein levels, thereby eliminating the differences between WT and Spp1-KO TAMs (Figures 5G and S5L). The downstream Stat1 activation and ISG expression also showed no further differences (Figures 5G and 5I). To determine whether Trim21 regulated Stat1 through its ubiquitination function, we treated IFN- γ -activated TAMs with the autophagy-lysosome inhibitor Bafilomycin A1 and the ubiquitin-proteasome inhibitor Bortezomib. We found Bafilomycin A1 had no impact on the Stat1 alterations caused by Spp1 deletion, whereas Bortezomib treatment abolished the differences in Stat1 (Figure 5J).

In summary, these findings suggest that SPP1 competes with SOCS1 for TRIM21 binding. Following SPP1 knockout, TRIM21-mediated binding and ubiquitination of SOCS1 increased, leading to reduced SOCS1 protein abundance and a consequent loss of negative regulation of the IFN- γ -STAT1-ISG pathway. In this context, SPP1 functions as a molecular shock absorber by buffering interferon signaling: it preserves SOCS1 stability and thereby maintains the feedback inhibition of IFN- γ -driven inflammatory responses. Loss of SPP1 removes this buffering effect, resulting in amplified interferon signaling.

IL-10 and TGF- β contribute to SPP1 expression

Our scRNA-seq analysis revealed that *Spp1* was among the most highly expressed genes in TAMs across multiple cancer types (Figure S6A). Additionally, our bulk RNA-seq data, complemented by qPCR validation of *in vitro*-induced TAMs, confirmed that *Spp1* expression levels exceed those of housekeeping genes *Gapdh* and *Actb* (Figures S6B and S6C). These observations prompted us to investigate the upstream signals that strongly induce SPP1 expression in TAMs. Given the immuno-

suppressive role of SPP1⁺ TAMs, we treated TAMs with common immunosuppressive factors to identify specific triggers of SPP1 expression. In addition to hypoxia, which is well known to induce SPP1 expression, we found that both interleukin (IL)-10 and TGF- β significantly stimulated SPP1 expression in TAMs (Figure S6D). Analysis of TCGA data further supported this finding, showing a significant positive correlation between *SPP1* expression and the levels of *IL-10* and *TGF- β* (Figures S6E and S6F). To better understand the underlying mechanisms, we treated IL-10-stimulated TAMs with Stattic, an inhibitor of an IL-10 downstream factor STAT3, and found that Stattic effectively suppressed IL-10-induced SPP1 overexpression (Figure S6G). Similarly, we treated TGF- β -activated TAMs with SIS3, an inhibitor of the activation of SMAD3 (a TGF- β downstream effector), and observed that SIS3 reduced TGF- β -induced SPP1 expression (Figure S6H).

We next tested *in vivo* whether inhibiting IL-10 and TGF- β signaling reduces SPP1 in TAMs and slows tumor progression. Consistent with a recent report,⁴³ pharmacologic inhibition of adenosine A2A receptors (Ciforadenant) suppressed tumor growth and TAMs-SPP1. By contrast, anti-IL-10R and Galunisertib (TGF- β pathway inhibitor) did not significantly affect tumor growth in this setting (Figures S6I and S6J). Galunisertib lowers TAMs-SPP1 yet fails to control tumors, likely because TGF- β receptors are broadly expressed across the TME (Figures S6K and S7A), which diminishes net benefit. For IL-10, although it induces SPP1 in macrophages *in vitro*, anti-IL-10R neither controlled tumors nor lowered TAMs-SPP1 *in vivo* (Figures S6J and S6K), consistent with low IL-10 expression in this model and indicating that TAMs-SPP1 is largely IL-10-independent here (Figures S7B and S7C). Analysis of human CRC scRNA-seq showed IL10/IL10R and TGFB/TGFBF expression patterns similar to those observed in our mouse datasets (Figures S7D and S7E). Thus, while TGF- β and IL-10 can induce SPP1 in macrophages *in vitro*, TGF- β blockade lacks cellular selectivity, and IL-10 blockade is context-dependent. Neither alone is an ideal strategy to lower TAMs-SPP1 for tumor control.

Targeting SPP1 augments ICB efficacy

As aforementioned, patients with elevated SPP1 in TAMs showed poor ICB response, while SPP1 ablation in TAMs

(B) qPCR tests measuring the expression levels of type I and type II interferon receptors in WT and Spp1-KO TAMs ($n = 3$ per group).

(C) qPCR tests measured indicated ISG expression levels in Spp1-KO TAMs cultured with complete growth media (10% FBS) alone and with recombinant Spp1 (rSpp1) at different concentrations after 24-h IFN- γ (25 ng/ μ l) treatment ($n = 3$ per group).

(D) qPCR tests measured indicated ISG expression levels in WT TAMs and Spp1-KO TAMs saved with mRNA transfection reagent only (Ctrl), sSpp1 isoform mRNA, or iSpp1 isoform mRNA after 48-h IFN- γ (25 ng/ μ l) treatment ($n = 3$ per group).

(E) Proteins precipitated with anti-flag nanobody from extracts of Spp1-KO TAMs transiently transfected with 3XFlag-tagged iSpp1 and analyzed with mass spectrometry. The top 15 proteins based on unique peptides and the sum of posterior error probability (PEP) scores are shown.

(F) Proteins precipitated from extracts of Spp1-KO TAMs transiently transfected with 3XFlag-tagged iSpp1 and immunoblotted for endogenous Socs1, Socs3, and Trim21. Immunoblots are representative of two independent experiments.

(G) Western blot analysis of the expression of Trim21, Socs1, Stat1, p-Stat1(S727), and p-Stat1(Y701) in WT and Spp1-KO TAMs after indicated treatment. The expression of the indicated protein was standardized against WT TAMs treated with IFN- γ (25 ng/ μ l). Using β -actin as a housekeeping protein. Immunoblots are representative of two independent experiments.

(H) qPCR tests measuring the expression levels of Socs1 and Socs3 in WT and Spp1-KO TAMs after indicated treatment ($n = 3$ per group).

(I) qPCR tests measured ISG expression levels in WT and Spp1-KO TAMs following Trim21 knockdown and subsequent treatment with IFN- γ ($n = 3$ per group).

(J) Western blot analysis of the expression of Stat1, p-Stat1(S727), and p-Stat1(Y701) in WT and Spp1-KO TAMs treated with PBS (Ctrl), autophagosome-lysosome inhibitor bafilomycin A1 (BafA1) (20 nM), and ubiquitin-proteasome inhibitor Bortezomib (2.5 nM) after being activated by IFN- γ (25 ng/ μ l). Immunoblots are representative of two independent experiments.

Data are shown as mean $\pm$ SEM, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$ by unpaired Student's t test, two-tailed (B, H, and I), or one-way ANOVA (C and D). See also Figures S5–S7.

increased immune cell infiltration and enhanced ISG and MHC class I signatures in malignant cells (Figures 1E–1G and 3A–3J). Consistently, analysis of pre- and post-ICB single-cell data across several tumor types revealed that higher ISG signatures were associated with improved ICB responses^{31–34} (Figures S8A–S8D). These results indicated that SPP1 knockout in TAMs might improve the response rates and overall efficacy of ICB therapy. We next assessed the antitumor effects of Spp1-KO TAMs plus anti-PD-L1 therapy in subcutaneous tumor models. Our experiments further demonstrated that SPP1 knockout in TAMs improved the effectiveness of PD-L1 therapy, resulting in a substantial tumor reduction, with complete regression observed in four out of seven cases (Figures 6A and 6B).

Furthermore, we observed that cancer cells mixed with Spp1-KO TAMs grew slower than those mixed with WT TAMs in our C57BL/6J Mix model (Figure 2I). Flow cytometry analysis revealed increased immune cell infiltration in the Spp1-KO group (Figure S9A). Concurrently, scRNA-seq analysis indicated a reduction in myeloid cell proportions, particularly macrophages and monocytes, accompanied by an increase in CD4⁺ T, CD8⁺ T, and NK cells (Figures S9B–S9D). Additionally, ISG expression and immune activation pathways were upregulated in the Spp1-KO group, suggesting that Spp1-KO TAMs can effectively induce a more immunogenic microenvironment compared with WT TAMs (Figures S9E and S9F). Thus, we hypothesized that using Spp1-KO macrophages for cell therapy might enhance ICB therapy. To validate our hypothesis, we injected cancer cells into the peritoneal cavity of mice, treating them with either WT or Spp1-KO BMDMs (Figure 6C). Spp1-KO BMDM treatment inhibited tumor progression and improved survival (Figures 6D and 6E). Additionally, combining Spp1-KO BMDMs with anti-PD-L1 therapy showed markedly greater efficacy than WT BMDMs (Figures 6D and 6E). Encouragingly, IFN- γ further amplified efficacy: after two rounds of combined IFN- γ , Spp1-KO BMDMs, and anti-PD-L1, complete tumor regression was achieved in 70% (7/10) of mice (Figures 6D, 6E, and S9G). These findings suggested that SPP1 ablation in macrophages enhances the efficacy of ICB, positioning SPP1 as a promising target for cell-based immunotherapy.

While macrophage transfer studies demonstrate that SPP1 ablation augments ICB, their clinical deployment remains limited. To strengthen translational relevance and directly validate SPP1 as a therapeutic target, we developed a gene-therapy strategy using lipid nanoparticle (LNP)-formulated small interfering RNA (siRNA). We first designed a panel of anti-SPP1 siRNAs and identified a top performer based on robust Spp1 knockdown in BMDMs *in vitro* (Figure S9H). We then introduced 2'-O-Me and 2'-F modifications with terminal phosphorothioate linkages to enhance stability, formulated the optimized siRNA in LNPs, and delivered it intratumorally, which inhibited tumor growth both as monotherapy and in combination with anti-PD-L1 (Figures 6F, 6G, S9I, and S9J). Together, these data support SPP1 in TAMs as a tractable target to potentiate ICB.

DISCUSSION

In this study, we showed the pro-tumor effects of SPP1⁺ TAMs by integrating large-scale single-cell transcriptomic data from multiple cancer types. Additionally, our previous studies re-

vealed that SPP1⁺ TAMs were linked with poor prognosis in patients of multiple cancer types, such as CRC.^{6,8,9} With an unsupervised ranking, we prioritized SPP1 as the leading marker of this subset, prompting the in-depth functional investigation of this gene in both tumor progression and immunotherapy. We demonstrated the potential of SPP1 as a target of TAMs-based immunotherapy by revealing that SPP1 ablation led to a favorable survival and improvement of anti-PD-L1 immunotherapy in multiple mouse CRC models. Further, we uncovered the cellular and molecular mechanism by which SPP1 deficiency in TAMs drives these anti-tumor effects.

TAMs facilitate tumor immune evasion by inhibiting T cell infiltration and/or impairing their cytotoxic functions.^{44–46} In this study, we observed that knocking out SPP1 in TAMs led to increased infiltration of immune cells, primarily lymphoid cells, including tumor-suppressive IFN- γ ⁺ CD4⁺ T cells and GZMB⁺ CD8⁺ T cells, together with elevated expression of ISGs, such as *Cxcl9*, *Cxcl10*, *Nos2*, and *Isg15* in TAMs. TAMs-derived CXCL9, CXCL10, and iNOS are known to play key roles in recruiting T cells, with CXCL9⁺ TAMs, CXCL10⁺ TAMs, and NOS2⁺ TAMs linked to T cell infiltration across various cancers.^{47–51} Additionally, secreted ISG15 has been shown to promote IFN- γ expression in both T cells and NK cells.⁵² In line with these findings, our data demonstrated that Spp1-KO TAMs, upon activation by IFN- γ , further enhanced IFN- γ expression in CD4⁺ T cells. This suggests a potential positive feedback mechanism linking Spp1-KO TAMs and IFN- γ ⁺ CD4⁺ T cells in immune regulation. In this loop, IFN- γ ⁺ CD4⁺ T cells secrete IFN- γ , activating Spp1-KO TAMs to increase their CXCL9, CXCL10, NOS2, and ISG15 expression. These cytokines, in turn, further enhance CD4⁺ T cell recruitment and differentiation of IFN- γ ⁺ CD4⁺ T cells, thereby fostering a more immunogenic tumor environment.

Mechanistically, we demonstrated that intracellular SPP1 participated in the negative feedback regulation of the IFN- γ -STAT1-ISG pathway in TAMs. While the IFN- γ -STAT1-ISG pathway is vital for immune defense, its hyperactivation can cause pathological consequences, making negative feedback crucial for maintaining immune balance.⁵³ SOCS1 serves as an important negative feedback regulator of the IFN- γ -STAT1-ISG pathway by inhibiting this signaling through targeting the upstream activators of STAT1, JAKs, either by preventing their activation or promoting their degradation.^{54,55} Additionally, SOCS1 may indirectly regulate STAT1 expression by inhibiting its activation and reducing IRF1's transcriptional output, as IRF1 binds to the STAT1 promoter and drives its transcription.^{56–58} This aligns with our findings that SPP1 knockout not only altered the activation of STAT1 but also resulted in significant changes in the expression of IRF1 and STAT1. Tripartite motif (TRIM) family proteins have been shown to regulate SOCS proteins with their E3-ubiquitin ligase activity. TRIM21 has been shown to interact with SOCS3 in thymic stromal cells.⁴² In this study, we demonstrated that intracellular SPP1 participates in the negative feedback regulation of the IFN- γ -STAT1-ISG pathway by competitively binding to TRIM21 with SOCS1. Notably, the efficiency of this regulatory mechanism may be influenced by the abundance of SPP1. Given that this function relies on SPP1 outcompeting SOCS1 for TRIM21 binding, its exceptionally high expression in TAMs across various cancer types—including breast cancer, melanoma, pancreatic

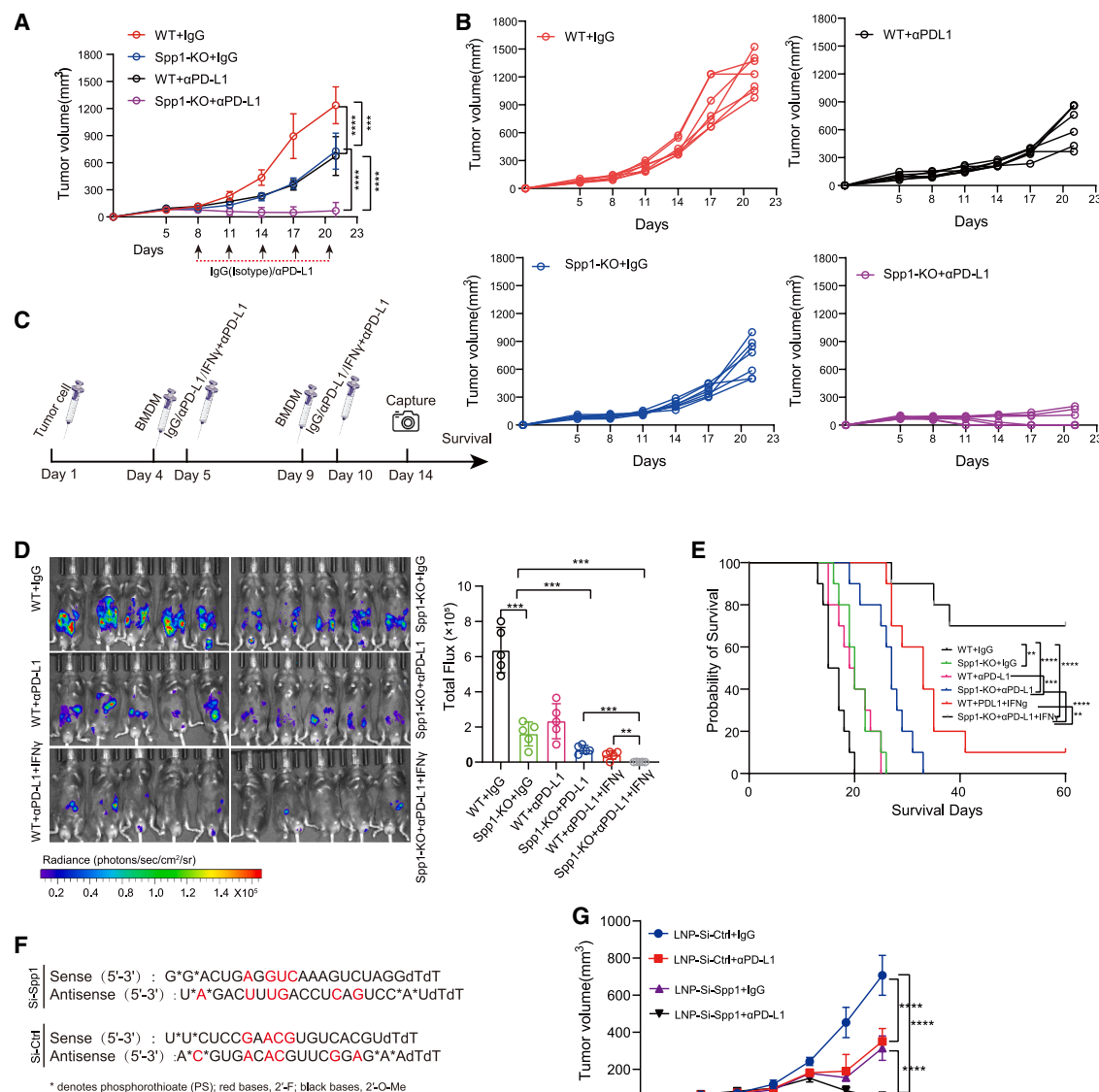


Figure 6. SPP1 knockout in TAMs enhances the efficacy of ICB therapy

(A and B) Tumor growth in mice injected subcutaneously with the MC38 cell line treated with IgG or anti-PD-L1 antibodies (A). Data are shown as a growth curve for each individual tumor (B) ($n = 6$ per group).

(C) Schematic diagram of the combined treatment plan.

(D and E) Bioluminescence analysis of the abdominal tumor burden (D) and survival analysis (E) of the indicated mice 14 days after intraperitoneal injection of 2×10^6 MC38-luciferase cells and indicated treatment. Survival analysis was performed using the log-rank (Mantel-Cox) test (tumor burden $n = 5$ per group; survival $n = 10$ per group).

(F) Schematic of siRNA sequences and modifications.

(G) Tumor growth curves after 1×10^6 MC38 cells subcutaneously transplanted into WT C57BL/6J under the indicated treatments ($n = 5$).

Data are shown as mean $\pm$ SEM; * $p < 0.05$, *** $p < 0.001$, **** $p < 0.0001$ by one-way ANOVA (A, D, and G).

See also Figures S8 and S9.

adenocarcinoma, and colon cancer—may be essential for enabling SPP1 to act as a “molecular shock absorber” that buffers excessive interferon signaling. This buffering capacity may be limited in contexts where SPP1 expression is low, such as in TAMs from myeloma or lymphatic cancers. Whether this mechanism operates in such settings remains to be determined.

We demonstrated that SPP1 knockout in BMDMs synergizes with anti-PD-L1 therapy, highlighting macrophage modulation as a promising strategy to enhance T cell-based therapies. Notably, in 70% of mice, complete tumor regression was observed following combined treatment with IFN- γ , Spp1-KO BMDMs, and anti-PD-L1 therapy, emphasizing the

therapeutic potential of targeting SPP1 in enhancing immune responses. This finding aligns with previous reports suggesting that modulating TAMs is a viable strategy to overcome immune resistance and enhance T cell-mediated anti-tumor effects.⁵ Despite the promising results, exploring effective strategies for targeting SPP1 necessitates further investigation. Although a few preclinical studies have suggested that anti-SPP1 antibodies may have therapeutic potential, clinical trials have shown no significant benefit.^{16,17,59,60} Our findings offer a potential explanation for this discrepancy: intracellular SPP1, which cannot be targeted by extracellular antibodies, plays a pivotal role in regulating the IFN- γ -STAT1-ISG pathway. Thus, antibody-based strategies may fail to recapitulate the full effects of SPP1 deletion in TAMs, particularly the intracellular functions that contribute to immune regulation and therapy resistance. To this end, we implemented an LNP-siRNA strategy that directly targets intracellular SPP1. When delivered intratumorally in LNPs, the siRNA inhibited tumor growth as monotherapy and enhanced anti-PD-L1 efficacy, thereby operationalizing SPP1 targeting in a drug-like, clinically oriented modality that captures the intracellular functions missed by antibody blockade. Moreover, chemical modifications enhance siRNA stability, and LNPs are efficiently taken up by macrophages, enabling sustained, macrophage-enriched SPP1 silencing and offering a promising avenue for future therapeutic development.⁶¹

In conclusion, our findings position SPP1 as a promising immunotherapy target, with the potential to convert immunosuppressive TAMs into pro-inflammatory players that facilitate anti-tumor immune responses. The results provide insights into the mechanisms underlying TAMs-mediated immune suppression and pave the way for novel combination therapies that could enhance ICB efficacy in solid tumors. Further research should focus on optimizing SPP1-targeted interventions and exploring their synergistic effects with existing immunotherapies in clinical settings.

Limitations of the study

Our therapeutic validation of SPP1 targeting relied on an LNP-formulated siRNA approach tested in immunocompetent C57BL/6J mice with syngeneic tumor models. While these experiments establish *in vivo* feasibility and efficacy in a physiologic immune setting, it remains to be determined whether comparable delivery, on-target silencing, and antitumor benefit are achieved in models that better approximate human tumor-immune interactions, including humanized mice and patient-derived xenograft (PDX) systems.

Mechanistically, the intracellular SPP1-TRIM21-SOCS1 framework was defined in tumor contexts in which TAMs express SPP1 at high levels, such as colorectal and lung cancer models. In other malignancies, including multiple myeloma and some lymphoid cancers, TAM SPP1 expression is often less prominent. Whether intracellular SPP1 similarly engages TRIM21 to modulate SOCS1 stability and constrain interferon responses in these settings, and more broadly, what the boundary conditions are for this regulatory axis, remains an important direction for future investigation.

RESOURCE AVAILABILITY

Lead contact

Requests for information and reagents should be directed to and will be fulfilled by the lead contact, Zemin Zhang (zemin@pku.edu.cn).

Materials availability

All unique reagents and data generated in this study are available from the [lead contact](#).

Data and code availability

Newly generated bulk RNA-seq and scRNA-seq data were deposited at Figshare: <https://doi.org/10.6084/m9.figshare.27628173>. This paper does not report original code. Additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

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AUTHOR CONTRIBUTIONS

L.S., X. Chu, L.Z., and Z.Z. contributed to conceptualization, writing of the original draft, and project administration. S.C., L.Z., and Z.Z. contributed to the review and editing of the paper. L.S., X. Chu, and X.W. contributed to software and/or code and formal data analysis. Q.G. and W.Z. performed R analysis and scRNA-seq. T.K., X. Chen, and J.R. were responsible for animal and *in vitro* experiments. Z.Z. was responsible for supervision and funding acquisition.

DECLARATION OF INTERESTS

Z.Z. is a founder of Analytical Bioscience Limited and also serves on the advisory board of Cell. X.W. is an employee of Analytical Biosciences Limited.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Rat APC anti-mouse CD45 Antibody	Biologend	Cat# 103112; RRID: AB_312977
FITC Rat anti-mouse CD45	BD Bioscience	Cat# 553079; RRID: AB_394609
BV510 Rat Anti-CD11B	BD Bioscience	Cat# 562950; RRID: AB_2737913
BUV496 Rat Anti-Mouse MHC class II	BD Bioscience	Cat#750281; RRID: AB_2874472
PerCP-Cy ^{5.5} Hamster Anti-Mouse CD11c	BD Bioscience	Cat#560584; RRID: AB_1727422
Brilliant Violet 605 [™] anti-mouse CD8a Antibody	BioLegend	Cat#100744; RRID: AB_2562609
PerCP-Cy ^{5.5} Rat Anti-Mouse CD4	BD Bioscience	Cat#550954; RRID: AB_393977
Alexa Fluor [®] 488 anti-mouse CD3 Antibody	Biologend	Cat#100210; RRID: AB_389301
APC anti-mouse IFN- γ Antibody	Biologend	Cat#505810; RRID: AB_315404
FITC anti-human/mouse Granzyme B Recombinant Antibody	Biologend	Cat#372206; RRID: AB_2687030
Alexa Fluor [®] 700 anti-mouse F4/80	Biologend	Cat#123130; RRID: AB_2293450
Alexa Fluor [®] 700 anti-mouse FOXP3 Antibody	Biologend	Cat#126422; RRID: AB_2750493
BV605 Rat Anti-Mouse CD25	BD Bioscience	Cat#563061; RRID: AB_2737982
PE/Cyanine7 anti-mouse NK-1.1 Antibody	Biologend	Cat# 108714; RRID: AB_389364
PE anti-mouse CD19 Antibody	Biologend	Cat#115508; RRID: AB_313643
Purified Rat Anti-Mouse CD16/CD32 (Mouse BD Fc Block [™])	BD Bioscience	Cat#553142; RRID: AB_394657
Anti- β -Actin (13E5) Rabbit mAb	CST	Cat#4970T; RRID: AB_2223172
Stat1 (D1K9Y) Rabbit mAb	CST	Cat#14994; RRID: AB_2737027
Phospho-Stat1 (Tyr701) (58D6) Rabbit mAb	CST	Cat#9167; RRID: AB_561284
Phospho-Stat1 (Ser727) Antibody	CST	Cat#9177; RRID: AB_2197983
Anti-FLAG [®] M2	CST	Cat# 70569; RRID: AB_2799786
SOCS1 (E3Q4M) Rabbit mAb	CST	Cat#55313; RRID: AB_3731289
Anti-SOCS3 antibody [EPR24090-74]	Abcam	Cat#ab280884; RRID: AB_3065200
Anti-TRIM21/SS-A antibody [EPR20290]	Abcam	Cat#ab207728; RRID: AB_2927717
CD4 (D7D2Z) Antibody	CST	Cat#25229; RRID: AB_2798898
CD45 (D3F8Q) Antibody	CST	Cat#70257; RRID: AB_2799780
CD11b (E4K8C) Antibody	CST	Cat#93169; RRID: AB_3731290
Ly-6G (E6Z1T) Antibody	CST	Cat#87048; RRID: AB_2909808
NK1.1 (E6Y9G) Antibody	CST	Cat#39197; RRID: AB_2943222
CD19 (D4V4B) Antibody	CST	Cat#90176; RRID: AB_2800152
CD8 α (D4W2Z) Antibody	CST	Cat#98941; RRID: AB_2756376
SPP1 Antibody	Abcam	Cat#ab216406; RRID: AB_3731292
CD3 Monoclonal Antibody (17A2)	Thermo Fisher	Cat#16-0032-82; RRID: AB_468851
CD28 Monoclonal Antibody (37.51)	Thermo Fisher	Cat#16-0281-82; RRID: AB_468921
Anti-mouse PD-L1 (B7-H1)-In Vivo	Selleck	Cat#A2115; RRID: AB_3675704
IgG2b isotype control-In Vivo	Selleck	Cat#A2116; RRID: AB_3662740
Anti-Mouse IL-10R Antibody (1B1.3A)	MCE	Cat#HY-P990228; RRID: AB_3731293
Rat IgG1 kappa, Isotype Control	MCE	Cat#HY-P99979; RRID: AB_3731294
Chemicals, peptides, and recombinant proteins		
DAPI Solution	BD Bioscience	Cat#564907
Red Cell Lysis Buffer	TIANGEN	Cat#RT122-02

(Continued on next page)

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Protein Transport Inhibitor	BD Bioscience	Cat#555029
BD Horizon™ Brilliant Stain Buffer	BD Bioscience	Cat#566349
Recombinant Mouse Osteopontin/OPN Protein	R&D	Cat#441-OP-050
Recombinant Osteopontin/OPN, Mouse (HEK293, His)	MCE	Cat#HY-P78358
Recombinant IFN-alpha 14, Mouse	MCE	Cat#HY-P76401
Recombinant IFN-beta, Mouse	MCE	Cat#HY-P73130
Recombinant IFN-gamma, Mouse	MCE	Cat#HY-P70667
Azoxymethane (AOM)	MP Biomedicals	Cat#2180139.1
Dextran Sodium Sulfate (DSS)	MP Biomedical	Cat#216011080
SIS3	SELLECK	Cat#S7959
Stattic	MCE	Cat#HY-13818
Galunisertib	MCE	Cat#HY-13226
Ciforadenant	MCE	Cat#HY-101978
Corn Oil	MCE	Cat#HY-Y1888
DMSO	SIGMA	Cat#D8418
Recombinant Murine IL-2	PEPROTECH	Cat#212-12
Recombinant Murine IL-4	PEPROTECH	Cat#214-14
Recombinant Murine IL-10	PEPROTECH	Cat#210-10
Recombinant Murine IL-12 p70	PEPROTECH	Cat#210-12
Recombinant IL-13, Mouse	MCE	Cat#HY-P70460
Recombinant TGF beta 1/TGFB1, Mouse	MCE	Cat#HY-P7117
Bafilomycin A1	MCE	Cat#HY-100558
Bortezomib	MCE	Cat#HY-10227
Recombinant Murine MCSF	PEPROTECH	Cat#315-02
D-Luciferin, Potassium Salt	Goldbio	Cat#LUCK-2G

Critical commercial assays

Zombie NIR™ Fixable Viability Kit	Biolegend	Cat#423106
Tumor Dissociation Kit, mouse	MILTENYI	Cat#130-096-730
EasySep™ Mouse CD4 ⁺ T Cell Isolation Kit	STEMCELL	Cat#19852
EasySep™ Mouse CD8 ⁺ T Cell Isolation Kit	STEMCELL	Cat#19853
RNeasy Mini Kit	QIAGEN	Cat# 74104
FOXP3/Transcription Factor Staining Buffer Set	Thermo Fisher	Cat# 00-5532-00
RNA isolater Total RNA Extraction Reagent	Vazyme	Cat# R401-01-AA
PrimeScript RT Reagent Kit	Takara	Cat# RR047A
TB Green Premix Ex Taq kit	Takara	Cat# RR420A
Anti-Flag Nanobody IP kit (Magarose)	AlpaLifeBio	Cat# KTSM1361
mMESSAGE mMACHINE™ T7 Transcription Kit	Thermo Fisher	Cat#AM1344
Lipofectamine™ MessengerMAX™ Transfection Reagent	Thermo Fisher	Cat#LMRNA001
HiPerFect Transfection Reagent	QIAGEN	Cat#301704
QIAquick Gel Extraction Kit	QIAGEN	Cat#28704
Anti-Flag Nanobody IP kit (Magarose)	AlpaLifeBio	Cat#KTSM1361
PANO 7-plex IHC kit	Panovue	Cat#10217100100

Deposited data

Processed Bulk RNA-seq data	This paper	https://doi.org/10.6084/m9.figshare.27628173
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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Processed single-cell RNA-seq data	This paper	https://doi.org/10.6084/m9.figshare.27628173
Monocyte and macrophage Single-cell transcriptomic data	Cheng et al. ⁶ Zhang et al. ⁹ Khosravi et al. ¹³ Kim et al. ¹⁴ Azizi et al. ¹⁸ Lambrechts et al. ¹⁹ Zilionis et al. ²⁰ Zhang et al. ²¹ Cillo et al. ²² Slyper et al. ²³ Lee et al. ²⁴ Qian et al. ²⁵ Zheng et al. ²⁶ Leader et al. ²⁷ Kim et al. ²⁸	GSE114727, E-MTAB-6149, E-MTAB-6653, GSE127465, EGAS00001003449, GSE139324, GSE123904, GSE146771, GSE131907, GSE140819, GSE132465, GSE132257, GSE144735, E-MTAB-8107, E-MTAB-6149, E-MTAB-6653, GSE145370, GSE154763, GSE154826, GSE150290
ICB treatment-related single-cell transcriptomic data	Chen et al. ⁷ Bi et al. ³¹ Krishna et al. ³² Shiao et al. ³³ Zhang et al. ³⁴	https://singlecell.broadinstitute.org/single_cell/study/SCP1288/tumor-and-immune-reprogramming-during-immunotherapy-in-advanced-renal-cell-carcinoma#study-summary , SRA: PRJNA705464, SRZ: SRZ190804, GSE169246, GSE246613, GSE236581
CRC single-cell atlas	Chu et al. ⁶²	https://doi.org/10.6084/m9.figshare.25323397
Experimental models: Cell lines		
MC38	Kerafast	Cat# ENH204
Luciferase-labeled MC38	This Paper	N/A
CT26	ATCC	Cat#CRL-2638
Experimental models: Organisms/strains		
Wild-type C57BL/6J	Shanghai Model Organisms	Cat#SM-001
NOD-SCID (NOD.Cg-Prkdc ^{scid} /NifdcSmoc)	Shanghai Model Organisms	Cat#SM-019
C57BL/6Smoc-Apc ^{tm2(flox)Smoc}	Shanghai Model Organisms	Cat#NM-CKO-200013
C57BL/6Smoc-Spp1 ^{em1(flox)Smoc}	Shanghai Model Organisms	Cat#NM-CKO-210205
Lyz2-Cre mice (B6.129P2-Lyz2 ^{tm1(cre)lfo/j})	Jackson Laboratory	Ca# 004781;
Vil1-MerCreMer mice	Cyagen	Cat#C001433
Oligonucleotides		
siRNA targeting Trim21: CAGAAUACCAAGAAGAGUACC	This study	N/A
List of primers	Table S3	N/A
Recombinant DNA		
pCDNA3.1(+)-intracellular Spp1-3xFlag	IGEBio	N/A
pCDNA3.1(+)-extracellular Spp1-3xFlag	IGEBio	N/A
pCDNA3.1(+)-Trim21-3xFlag	IGEBio	N/A
Software and algorithms		
FlowJo 10.3	BD Biosciences	https://www.flowjo.com
FIJI (ImageJ)	ImageJ	https://fiji.sc/
Imaris v.9.3.1	Oxford Instruments plc	https://imaris.oxinst.com/
GraphPad Prism Version 9	GraphPad Prism	www.graphpad.com
TCGAbiolinks (v2.22.4)	Bioconductor	https://bioconductor.org/packages/release/bioc/html/TCGAbiolinks.html
Seurat (v5.0.1)	Hao et al. ⁶³	https://satijalab.org/seurat/index.html
Harmony (v1.1.0)	Github	https://github.com/immunogenomics/harmony
CellRanger v6.12	10x Genomics	https://www.10xgenomics.com/
fastp (v0.18.0)	Github	https://github.com/OpenGene/fastp
HISAT2 (v2.4)	Github	http://daehwankimlab.github.io/hisat2/

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
StringTie (v1.3.1)	Github	https://github.com/gpertea/stringtie
RSEM (v1)	Github	https://github.com/deweylab/RSEM
DESeq2 (v1.42.0)	Bioconductor	https://bioconductor.org/packages/release/bioc/html/DESeq2.html
clusterProfiler (v4.10.0)	Bioconductor	https://www.bioconductor.org/packages/release/bioc/html/clusterProfiler.html
GSVA (v1.50.0)	Bioconductor	https://www.bioconductor.org/packages/release/bioc/html/GSVA.html
SingleR (v2.4.1)	Github	https://github.com/dviraran/SingleR
CellChat (v1.6.1)	Github	https://github.com/sqjin/CellChat
ggplot2 (v3.4.2)	CRAN	https://cran.microsoft.com/web/packages/ggplot2/
pheatmap (v1.0.12)	CRAN	https://cran.r-project.org/web/packages/pheatmap/
VennDiagram (v1.7.3)	CRAN	https://cran.r-project.org/web/packages/VennDiagram/index.html
Other		
N/A	N/A	N/A

METHOD DETAILS

Animals

All animal experiments in this study were reviewed and approved by the Institutional Animal Care and Use Committee of Shenzhen Bay Laboratory (AEZZM20201). Wild-type C57BL/6J mice, NOD-SCID mice (NOD.Cg-Prkdc^{scid}/NifdcSmoc), Apc-flox mice (C57BL/6Smoc-Apc^{tm2(flox)Smoc}), and Spp1-flox mice (C57BL/6Smoc-Spp1^{em1(flox)Smoc}) were purchased from the Shanghai Model Organisms Center, Inc (Shanghai, China). Vil1-Cre (Vil1-MerCreMer) mice were obtained from Cyagen Biosciences (Guangzhou, China). Lyz2-Cre mice (B6.129P2-Lyz2^{tm1(Cre)fl}) were purchased from Jackson Lab. Mice were bred under SPF conditions, maintained on a 12-hour light/dark cycle at 21–22°C with 30–70% humidity. Mice within experiments were age and sex matched and all mice were genotyped before the experiment.

Tumor models

To generate macrophage-specific Spp1 knockout mice, we crossed Spp1-flox mice with Lyz2-Cre mice. Spp1^{wt/wt} Lyz2-Cre^{+/-} mice were designated as WT and Spp1^{flox/flox} Lyz2-Cre^{+/-} mice were designated as Spp1-KO. To establish a subcutaneous tumor model, we resuspend MC38/LLC1 cells in a mixture of PBS and Matrigel (Corning, 354234) (1:1) at a concentration of 1×10⁷ cells/ml. WT and Spp1-KO male mice at 6 weeks of age were used and injected with 200 μL MC38 (or 100 μL LLC1) cell suspension subcutaneously into the flank region.

For the macrophage depletion experiment, mice received clodronate liposomes by i.p. injection (200 μL/mouse) 1 day before tumor inoculation, then every 3 days thereafter. On day 0, 1 × 10⁶ MC38 cells in 100 μL mixture of PBS and Matrigel were injected subcutaneously into the right flank to establish tumors. Tumors were measured with a caliper 5 days after injection and then every three days thereafter.

For the C57BL/6J Mix tumor model, we mixed MC38 cells and WT or Spp1-KO BMDMs (1:1) and resuspended cells in a mixture of PBS and Matrigel (1:1) at a concentration of 2×10⁷ cells/ml. Male C57BL/6J mice at 6 weeks of age were used and injected with 200 μL cell suspension subcutaneously into the flank region. The NOD-SCID Mix tumor model was constructed using the same method as the C57BL/6J Mix tumor model mentioned above. Tumors were measured with a caliper 5 days after injection and then every three days thereafter. Mice were sacrificed when the biggest tumor reached 1.5 cm in diameter.

To establish a peritoneal tumor model, we resuspend the luciferase-labeled MC38 cells in cold PBS at a concentration of 1×10⁷ cells/ml and injected with 200 μL cell suspension intraperitoneally (IP) by inserting the needle into the lower right quadrant of the abdomen. The tumor burden was assessed by in vivo bioluminescence at 14 days after injection. As for survival assays, tumor-bearing mice were monitored every two days for signs of disease progression or death. Humane endpoints are defined by a veterinarian and euthanized by CO₂ asphyxiation.

The bone marrow chimeric APC/DSS spontaneous colorectal cancer model was conducted as previously described.³⁷ Briefly, we crossed Apc-flox mice with Vil1-Cre mice to generate Apc^{flox/wt}Vil1-Cre^{+/-} mice. At 6 weeks old, Apc^{flox/wt}Vil1-Cre^{+/-} mice were given a split dose (~4 hours apart) of lethal irradiation totaling 10.5 Gy prior to transplantation via retro-orbital injection of 2 × 10⁶ unfractionated whole bone marrow cells harvested from 8-week-old WT or Spp1-KO mice. At 8 weeks old, bone marrow chimeric

Apc^{fllox/WT}Vil1-Cre^{+/-} mice were treated with 2% dextran sodium sulfate (DSS, MP Biomedical, 216011080) in drinking water for 1 week then replaced with regular drinking water for two weeks. And repeated DSS treatment three times to enhance colon tumor formation. Mice were euthanized 2 weeks later, and the number and size of tumors in the colon were recorded. For survival assays, tumor-bearing mice were monitored every two days for signs of disease progression or mortality. Humane endpoints were determined by a veterinarian, and euthanasia was performed using CO₂ asphyxiation.

In vivo treatments

For *in vivo* validation of IL-10/TGF- β blockade, Wild-type C57BL/6J mice were implanted subcutaneously with 1×10^6 MC38 cells. Once tumors were measurable, mice were randomized ($n = 5/\text{group}$) to the indicated treatments: Ciforadenant (10 mg/kg, oral gavage), Galunisertib (75 mg/kg, oral gavage), or vehicle (10% DMSO / 90% corn oil, oral gavage). For IL-10 blockade, anti-IL-10R (CD210) or isotype IgG control (100 $\mu\text{g}/\text{mouse}$) was administered by peritumoral injection. Tumor dimensions were recorded every 3 days. At endpoint, tumors were dissociated, TAMs (CD45⁺CD11b⁺F4/80⁺) were sorted by flow cytometry, and Spp1 mRNA was quantified by qPCR.

In the subcutaneous tumor model, para-tumor injections of control IgG (100 $\mu\text{g}/\text{mouse}$, Selleck, A2116) or anti-PD-L1 antibody (100 $\mu\text{g}/\text{mouse}$, Selleck, A2115) every 3 days, starting at day 8 after tumor injection. In the peritoneal tumor model, tumor-bearing mice were treated intraperitoneally (i.p.) with WT or Spp1-KO BMDMs starting on day 4 after cancer cell injection. One day later, mice were treated with IgG, anti-PD-L1 antibody (100 $\mu\text{g}/\text{mouse}$), or a combination of either IgG or anti-PD-L1 antibody with IFN γ (200 ng/mouse, MCE, HY-P70667). This treatment regimen was repeated starting on day 9. The tumor burden was assessed by *in vivo* bioluminescence at 14 days by intraperitoneal injection of D-Luciferin potassium (150 mg/kg, Goldbio, LUCK-2G).

The subcutaneous implantation and group allocation for the *in vivo* LNP-siRNA study followed the same protocol as the antibody-treatment experiments. The siRNA (si-SPP1 or si-Ctrl) was formulated into MC3-based LNPs (DLin-MC3-DMA: Chol: DSPC: DMG-PEG2000 $\approx 50 : 38.5 : 10 : 1.5$ mol%; mean diameter $\sim 70\text{--}90$ nm; PDI < 0.2) and administered by intratumoral injection at 23 μg per dose, every 2 days. For combination groups, anti-PD-L1 (or IgG control) was co-administered by intratumoral injection at 100 μg per mouse, once every 3 days, alongside the LNP-siRNA regimen.

Cell culture

MC38 was purchased from Kerafast (Boston, MA; ENH204). CT26 was obtained from ATCC (Manassas, VA; CRL-2638). The above cell lines were grown in DMEM (Gibco), supplemented with 10% fetal bovine serum (Gibco) and a 1% penicillin/streptomycin mixture (Gibco), and maintained at 37 $^{\circ}\text{C}$ in a humidified atmosphere containing 5% CO₂.

We euthanized the mouse and extracted the femur and tibia to derive bone marrow-derived macrophages (BMDMs) from mice. The bone marrow was flushed out using a syringe filled with cold PBS and filtered through a 40 μm strainer. The cells were then centrifuged and resuspended in RPMI-1640 medium supplemented with 10% FBS, 1% penicillin-streptomycin (PS), and 25 ng/mL M-CSF. The cells were incubated for 6 days, with media replaced every 3 days to promote differentiation into macrophages. On day 7, the adherent macrophages were gently harvested by scraping and co-cultured with tumor cells for 48 hours to induce their differentiation into TAMs. Before the subsequent experiments, the phenotype of TAMs was confirmed through flow cytometry. For hypoxic cell culture, TAMs were exposed to 1% O₂ for 24 hours prior to harvesting.

In vitro T Cells polarization

For non-IFN γ activated assays, sorting the *in vitro*-induced WT and Spp1-KO TAMs and seeding them into a 96-well plate at 2000 cells per well for 48 hours of culture. For IFN γ activated assays, seeding them into a 96-well plate at 2000 cells per well and treating *in vitro*-induced WT and Spp1-KO TAMs with 25 ng/mL IFN γ for 48 hours. Refresh the medium before the co-culture experiment. Naive CD4⁺ and CD8⁺ T cells were isolated from the 6 weeks old wild-type C57BL/6J mouse spleen using a naive T cell isolation kit, following the manufacturer's protocol (BioLegend, 480039, 480043). For GZMB⁺CD8⁺T cells, resuspend naive CD8⁺T cells in RPMI-1640 medium containing 10% FBS, 1% PS, soluble antibody against CD3 (3 $\mu\text{g}/\text{ml}$, Thermo Fisher, 16-0032-82), soluble antibody against CD28 (3 $\mu\text{g}/\text{ml}$, Thermo Fisher, 16-0281-82), IL-2 (20 U/ml, PEPROTECH, 212-12), and IL-12 (20 ng/ml, PEPROTECH, 210-12) at a concentration of 2×10^5 cells/ml. For Treg cells, resuspend naive CD4⁺T cells in RPMI-1640 medium containing 10% FBS, 1% PS, soluble antibody against CD3 (3 $\mu\text{g}/\text{ml}$), soluble antibody against CD28 (3 $\mu\text{g}/\text{ml}$), IL-2 (20 U/mL), and recombinant TGF β (5 ng/ml, MCE, HY-P7117) at a concentration of 2×10^5 cells/ml. For IFN γ ⁺CD4⁺T cells, resuspend naive CD4⁺T cells in RPMI-1640 medium containing 10% FBS, 1% PS, soluble antibody against CD3 (3 $\mu\text{g}/\text{ml}$), soluble antibody against CD28 (3 $\mu\text{g}/\text{ml}$), IL-2 (20 U/mL), and IL-12 (20 ng/mL) at a concentration of 2×10^5 cells/ml. For T cell proliferation, resuspend naive CD4⁺T/CD8⁺T cells in RPMI-1640 medium containing 10% FBS, 1% PS, soluble antibody against CD3 (1.5 $\mu\text{g}/\text{ml}$), soluble antibody against CD28 (1.5 $\mu\text{g}/\text{ml}$), IL-2 (10 U/mL), at a concentration of 2×10^5 cells/ml. Transfer 100 μL of the above cell suspension into a 96-well plate coated with IFN γ activated or non-activated TAMs and culture for 2.5 days. Assessing T cell polarization using flow cytometry.

T-cell recruitment

WT or Spp1-KO TAMs were seeded in the lower chamber and allowed to condition the medium for 48 h. Purified CD4⁺ or CD8⁺ T cells were then added to the upper insert. After 5 h, T cells that migrated to the lower chamber were collected and enumerated, and migration was expressed as the percentage of migrated T cells per condition.

Flow Cytometry, staining, and gating strategies

Tumors were minced and digested using the Tumor Dissociation Kit (Miltenyi, 30-096-730) according to the manufacturer's instructions. The resulting single-cell suspensions were filtered through 70- μ m strainers, washed with cold PBS, and incubated with Fc blocking antibody (CD16/32, BD Biosciences, 553142) for 10 minutes at 4°C. Cells were then stained with a cocktail of fluorochrome-conjugated antibodies for 30 minutes at 4°C. For cell surface staining, DAPI was used to exclude dead cells. For intracellular staining, dead cells were labeled using the Zombie NIR Fixable Viability Kit (BioLegend, 423105) prior to fixation and staining. The Foxp3/Transcription Factor Staining Buffer Set (Thermo Fisher, 00-5532-00) was used for subsequent staining procedures. To detect intracellular IFN γ and GZMB, cell suspensions were incubated in a 96-well plate with Brefeldin A (BioLegend) for 3 hours at 37°C before continuing with staining as described. Cells were analyzed on the CytoFLEX LX (Beckman), and sorting was performed using the FACSARIA II (BD Biosciences).

The following gating strategies were applied for the analysis of mouse tumor samples: Total immune cells: DAPI⁻CD45⁺; CD4⁺T cells: DAPI⁻CD45⁺CD3⁺CD4⁺; CD8⁺T cells: DAPI⁻CD45⁺CD3⁺CD8⁺; NK cells: DAPI⁻CD45⁺CD3⁺NK1.1⁺; B cells: DAPI⁻CD45⁺CD19⁺; TAMs: DAPI⁻CD45⁺CD11b⁺F4/80⁺; DCs: DAPI⁻CD45⁺CD11c⁺MHCII⁺; Myeloid cells: DAPI⁻CD45⁺CD11b⁺; NIR⁻CD45⁺CD3⁺CD4⁺IFN γ ⁺ was used to detect IFN γ in CD4⁺T cells, while NIR⁻CD45⁺CD3⁺CD4⁺CD25⁺FOXP3⁺ was used to identify Tregs, and DAPI⁻CD45⁺CD3⁺CD8⁺GZMB⁺ was used to detect GZMB in CD8⁺T cells. For the in vitro TAMs and T cell co-culture model, the following gating strategies were applied: NIR⁻CD4⁺CD25⁺FOXP3⁺ for Tregs, NIR⁻CD4⁺IFN γ ⁺ for IFN γ expression in CD4⁺T cells, and NIR⁻CD8⁺GZMB⁺ for GZMB expression in CD8⁺T cells. A complete list of antibodies and reagents used in this study is provided in the KEY RESOURCES TABLE.

Immunofluorescence staining

Tumor specimens were fixed in 4% paraformaldehyde (PFA) at 4°C overnight with agitation, then transferred to 70% ethanol for preservation and subsequently embedded in paraffin. Sections were blocked for 1 hour in 5% BSA and stained using a multiplex immunofluorescence PANO 7-plex IHC kit (Panovue, 10217100100) according to the manufacturer's instructions. CD45 (D3F8Q) (1:200, CST, 70257), CD11b (E4K8C) (1:50, CST, 93169), Ly-6G (E6Z1T) (1:100, CST, 87048), NK1.1 (E6Y9G) (1:50, CST, 39197), CD19 (D4V4B) (1:1600, CST, 90176), CD8 α (D4W2Z) (1:200, CST, 98941), CD4 (D7D2Z) (1:50, CST, 25229), SPP1 (1:100, Abcam, ab216406). DAPI staining was performed at room temperature for 10 minutes. The sections were then mounted using ProLong Diamond anti-fade mounting media (Thermo Fisher, P36961). Data acquisition was performed using the VS200 slide scanner, and image analysis was carried out with Imaris v.9.3.1 and ImageJ software.

Gene expression quantification by real-time PCR

Total RNA was extracted using RNA isolater Total RNA Extraction Reagent (Vazyme, R401-01-AA), followed by cDNA synthesis through reverse transcription with the PrimeScript RT Reagent Kit and gDNA Eraser (Takara, RR047A). Quantitative PCR (qPCR) was conducted using the TB Green Premix Ex Taq kit (Takara, RR420A), with Ct values detected by the CFX384 Touch™ Real-Time PCR Detection System (BioRad). Relative mRNA expression levels were normalized to the internal reference gene 18S. The primer sequences are listed in Table S3.

In vitro coculture and bulk RNA-seq

In vitro-induced BMDMs were scraped and co-cultured with MC38 tumor cells at a 1:1 ratio for two days. Following co-culture, cells were harvested and stained with anti-CD45 antibody and DAPI. Tumor cells (DAPI⁻CD45⁻) and TAMs (DAPI⁻CD45⁺) were subsequently sorted, with 2x10⁶ cells from each population collected for RNA sequencing. Total RNA was isolated using Total RNA Extraction Reagent (Vazyme, R401-01-AA), and full-length cDNA was synthesized using the SMART-Seq HT Kit (Takara; 634436). Libraries were constructed using the Nextera XT DNA Library Preparation Kit (Illumina, San Diego, CA, USA), and sequencing was conducted on the Illumina NovaSeq 6000 platform in 100-base single-end mode.

Western blots

WT/Spp1-KO TAMs were treated with 25 ng/ml IFNs for 48 hours and 2x10⁶ cells were lysed in RIPA Lysis and Extraction Buffer (Thermo Scientific, 89901) supplemented with protease and phosphatase inhibitors, and protein concentrations were quantified using the Pierce BCA Protein Assay Kit.

To determine whether TRIM21 regulated STAT1 through its ubiquitination function, we treated IFN γ -activated TAMs with 20 nM autophagy-lysosome inhibitor Bafilomycin A1 or 2.5 nM ubiquitin-proteasome inhibitor Bortezomib for 48 hours. The protein samples were resolved by SDS-PAGE and transferred onto polyvinylidene fluoride (PVDF) membranes (Roche). Membranes were blocked in 5% BSA and incubated overnight at 4°C with primary antibodies. The following primary antibodies were used: Anti- β -actin (CST, 4970T, 1:1000), Anti-Stat1 (CST, 14994s, 1:1000), Anti-p-Stat1(S727) (CST, 9177s, 1:1000), Anti-p-Stat1(Y701) (CST, 9167s, 1:1000), anti-Socs1 (CST, 55313S, 1:1000), anti-Socs3 (Abcam, ab280884, 1:1000), anti-Trim21 (Abcam, ab207728, 1:1000) and anti-Flag (CST, 55313S, 1:1000). After TBST washing, membranes were incubated with HRP-conjugated secondary antibodies (CST, 7074S, 1:5000), and signals were visualized using Clarity Western ECL Substrate (Bio-Rad). Western blot data were analyzed using ImageJ software. All raw western blot images are provided in Figure S10.

Ectopic gene expression in macrophages using in vitro transcribed mRNA

pCDNA3.1(+)-intracellular Spp1-3xFlag, pCDNA3.1(+)-extracellular Spp1-3xFlag and pCDNA3.1(+)-Trim21-3xFlag plasmids were generated as PCR templates (IGEBio). We designed forward and reverse primers specific to the flag-tagged intracellular Spp1, secreted Spp1 and Trim21 mRNA, appending the T7 RNA polymerase promoter sequence (T7) [5'-AGTAATACGACTCACTATAGGG-3'] upstream of the forward primer. The T7 DNA templates were amplified following the instructions of SMART-Seq HT Kit (Takara; 634436), and the PCR products were purified using the QIAquick Gel Extraction Kit (QIAGEN; 28704) according to the manufacturer's instructions. The mRNA fragments were then transcribed in vitro using the mMESSAGE mMACHINE™ T7 Transcription Kit according to the manufacturer's protocol (Thermo Scientific, AM1344). The in vitro transcribed mRNA fragments were purified and quality-checked. Next, transfect 1 μg intracellular Spp1-3xFlag, secreted Spp1-3xFlag or Trim21-3xFlag mRNA into 2x10⁶ Spp1-KO or WT TAMs using the Lipofectamine™ MessengerMAX™ Transfection Reagent (Thermo Scientific, LMRNA003).

Co-IP assay

Spp1-KO TAMs were transfected with intracellular Spp1-3XFlag mRNA, and 24 hours post-transfection, the cells were treated with 25 ng/mL of IFN γ to activate the TAMs for an additional 48 hours. Following activation, 1x10⁸ TAMs were washed with cold PBS and lysed using lysis buffer. To avoid interference from the light and heavy chains, co-immunoprecipitation was carried out using Anti-Flag Nanobody IP kit (Magarose) (AlpaLifeBio, KTSM1361), following the manufacturer's instructions. The precipitated proteins were subsequently utilized for Western blot or mass spectrometry analysis.

LC-MS/MS analysis

Proteins immunoprecipitated using Anti-Flag and negative control beads were digested with trypsin for 16 hours at 37°C with agitation at 600 rpm in darkness. After desalting, the resulting peptides were lyophilized and analyzed by ultra-high-performance liquid chromatography (Acquity, Waters) coupled to a Q Exactive Plus Hybrid Quadrupole-Orbitrap Mass Spectrometer (Thermo Fisher). Data analysis was performed using Proteome Discoverer software.

Foci formation

Seeded 1.5 × 10⁶ WT or Spp1-KO TAMs into 6-well plates and cultured them for two days. After incubation, collected the supernatant and diluted it 1:1 with fresh DMEM medium containing 10% FBS and 1% PS to prepare the TAMs-conditioned medium. Next, seeded 500 MC38 or CT26 cells per well into 6-well plates and added 2 mL of TAMs-conditioned medium to each well. Replaced the WT or Spp1-KO TAMs-conditioned medium every two days. On day eight, fixed and stained the cells with crystal violet, then quantified the colony numbers.

siRNA-mediated silencing

TAMs were transfected with 25 nM siRNA for 24 hours using HiPerFect Transfection Reagent (QIAGEN, 301704), following the manufacturer's instructions. The cells were then stimulated with 25 ng/mL IFNs for 48 hours prior to PCR or Western blot analysis.

Human scRNAseq data collection and quality control

Monocyte and macrophage Single-cell transcriptomic data of 12 cancer types from 15 studies,^{6,9,13,14,18–28} covering PBMC, adjacent normal tissues and tumors, were collected and quality controlled by removing cells expressing less than 800 genes. ICB treatment-related single-cell transcriptomic data were acquired from public studies.^{7,31–34} Processed expression matrices of TAMs were extracted for analyses. The CRC single-cell atlas was obtained from our previously published data.⁶²

TCGA data collection

Gene expression matrix in TPM format of each cancer type was acquired from TCGA database (<https://portal.gdc.cancer.gov/>) using the R package TCGAbiolinks (v2.22.4) in December 2022.

Clustering and cell subset identification in human scRNAseq data

For constructing the monocyte and macrophage single-cell atlas, an R package Seurat (v5.0.1)⁶³ was applied first for normalization and highly variable gene identification. The top 2000 variable genes excluding mitochondrial genes were next scaled for principal component analysis (PCA). We corrected the top 30 principal components (PCs) by different datasets using Harmony (v1.1.0)⁶⁴ and the corrected PCs were used for cell clustering by constructing a K nearest neighbor (KNN) graph with default settings and resolution = 0.4 and for visualization with a Uniform Manifold Approximation and Projection (UMAP) approach. Marker genes for each cell subset were identified using Wilcoxon rank-sum tests on the genes expressed in >25% of the cells. P values of multiple tests were corrected using the *Benjamini-Hochberg* approach and genes with adjusted p value less than 0.05 and log2 fold change larger than 0.5 were considered to be upregulated genes.

Tissue preference evaluation of monocyte and macrophage subsets

To detect the tissue preference of each monocyte and macrophage subset, a ratio of observed/expected (*Ro/e*) cell numbers was calculated. The expected cell numbers were obtained from the chi-square test, as previously described.^{29,30}

Mouse single-cell transcriptome sequencing

To examine changes in the tumor microenvironment following SPP1 knockout in TAMs, DAPI⁺CD45⁺ and DAPI⁺CD45⁻ cells were sorted from MC38 subcutaneous tumors. After staining with Trypan blue, cells were counted, and 10,000 live cells were loaded onto a Chromium Chip A (10X Genomics, PN 230027). GEM generation, cDNA synthesis, amplification, and library preparation were conducted using the Chromium Single Cell 5' Reagent Kit (10X Genomics, PN 1000006), following the manufacturer's protocol. Indexed libraries were pooled in equimolar ratios and sequenced on a NextSeq 500 in a 26 bp/91 bp paired-end run with the NextSeq 500/550 High Output Kit v2.5 (150 cycles, Illumina).

Mouse models scRNAseq data preprocessing and quality control

CellRanger v6.12 (10x Genomics) was applied to process raw sequencing data with default settings and reads from single cells were aligned to the mm10 reference genome. Next, Seurat was applied to analyze the gene expression matrix. To acquire an accurate cell type annotation, for conditional Spp1-KO data, CD45⁺ cells with less than 800 detected genes and with more than 20% mitochondrial counts were excluded and for the mix model data, CD45⁺ cells with less than 500 detected genes and with more than 10% mitochondrial counts were excluded. The potential doublets, which co-expressed different well-known cell type marker genes, were also filtered out, leaving 17659 cells for conditional Spp1-KO data and 12298 cells for the mix model. CD45⁻ cells of the conditional Spp1-KO data expressing less than 2000 genes and more than 20% mitochondrial counts were excluded, ending up with 15275 cells in the downstream analysis.

Clustering and cell subset identification in mouse models' scRNAseq data

By applying Seurat, we clustered and annotated cell subsets following the procedure below. First, we normalized the count matrix and identified the top 2000 variable genes, the expression levels of which were next scaled and used for PCA. The first 20 PCs were used to construct KNN graph with default settings and resolutions ranging from 0.5 to 1 were selected for cell clustering. For visualization, UMAP was implemented on the first 20 PCs. Finally, cell types were identified based on the expression pattern of canonical markers and over-presentative genes in each cell cluster. We first went through the procedure to identify major cell types in CD45⁺ cells and CD45⁻ cells respectively. Then we followed the procedure again to identify cell subsets within myeloid cells and lymphocytes of CD45⁺ cells respectively. Marker genes for each cell subset were identified using Wilcoxon rank-sum tests on the genes expressed in >25% of the cells. P values of multiple tests were corrected using the *Benjamini-Hochberg* approach and genes with adjusted p value less than 0.05 and log2 fold change larger than 0.5 were considered to be upregulated genes.

Differential expression analysis in mouse models of scRNA-seq data

To identify differentially expressed genes between different treatment groups, we applied Wilcoxon's tests using the function "FindMarkers" from Seurat on genes expressed in >25% of the cells from any group, and P values of multiple tests were corrected using the *Benjamini-Hochberg* approach. Genes with adjusted p value less than 0.05 and log2 fold change larger than 0.5 were considered to be differentially expressed genes.

Bulk RNA-seq data analysis

Sequencing reads were first filtered by fastp (v0.18.0)⁶⁵ to remove reads containing adapters, containing more than 10% unknown nucleotides, or containing more than 50% of low-quality (Q-value ≤ 20) bases. Next, HISAT2 (v2.4) was applied for alignment to the mouse reference genome (Ensembl_release107). The mapped reads were assembled and quantified using StringTie (v1.3.1)⁶⁶ and RSEM (v1)⁶⁷ respectively. To perform differential expression analysis, an R package DESeq2 (v1.42.0)⁶⁸ was applied, and the genes with FDR less than 0.05 and absolute fold change ≥ 2 were considered differentially expressed.

Gene ontology (GO) and Hallmark term enrichment analysis

To identify over-represented GO biological processes and Hallmark terms, an R package clusterProfiler v4.10.0 was applied and p values adjusted with the *Benjamini-Hochberg* approach less than 0.05 were considered statistically significant.

Gene set variation analysis (GSVA) and scoring of signature genes

To estimate the variation of GO or Hallmark terms enrichment in scRNAseq data, an R package GSVA (v1.50.0) was applied with the method "ssgsva". The scores of different groups of cells for each term were next compared using two-sided unpaired t-tests with p values corrected using the *Benjamini-Hochberg* approach, and FDR values less than 0.05 were considered statistically significant. To evaluate the signature of interesting cell subsets and biological pathways, the signature gene lists were acquired from previous studies^{9,69,70} and an "AddModuleScore" function from Seurat was implemented on the expression values of each cell.

Macrophage type inference

To estimate the similarity of TAMs to M1 macrophages and M2 macrophages, we downloaded the single-cell transcriptomic data of in vitro induced M1, M2 and unpolarized macrophages at 24h after treatments from a public study.⁴¹ The top 50 marker genes with the largest log2 fold changes were first identified using the "Find All Markers" function from Seurat. Then an R package SingleR (v2.4.1)⁷¹ was applied taking this data as the reference to annotate each TAMs to either M1, M2, or unpolarized macrophage with only max prediction scores > 0.3 were kept for downstream analysis.

Cell-cell interaction inference

We used Cell Chat (v1.6.1)⁷² to identify cell-cell interactions between cell types with more than 50 cells for each group with default settings.

Other visualization

The R package ggplot2 (v3.4.2) and Seurat were applied for most visualizations. The package pheatmap (v1.0.12) was used to generate most heatmaps and the package Venn Diagram was used to generate the Venn diagram (v1.7.3).

STATISTICAL ANALYSIS

For statistical analysis, Student's t-test (two-tailed) was used for experiments with only two groups. For experiments with more than two groups, one-way ANOVA with multiple comparisons was performed. Survival analysis was conducted using the Kaplan–Meier method and the log-rank test. Data are expressed as mean ± SEM, with statistical significance defined as * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, and **** $p < 0.0001$.